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Global burden of disease and its risk factors for adults aged 70 and older across 204 countries and territories: a comprehensive analysis of the Global Burden of Disease Study 2021

Qin-Fen Chen[1,2†], Chao Ni[2†], Yalan Jiang[2], Lifen Chen[2], Hetong Liao[2], Jiaqi Gao[2], Xuemei Qin[2], Sipei Pan[2], Xiaoqian Luan[2], Yili Wu[2,3*], Xiao-Dong Zhou[4*] and Weihong Song[2,3*]

Abstract

Background The global older population has been rapidly increasing in recent decades, presenting substantial economic and social challenges.

Aims This study aims to examine the global disease burden and the associated risk factors among the population aged 70 and older across different regions.

Methods Data from the Global Burden of Disease Study (GBD) database were used to analyze global deaths, disability-adjusted life years (DALYs), and average annual percentage changes to assess disease burden and associated risk factors.

Results In 2021, the global population aged 70 and older reached 494.4 (95%UI 482.1–508.2) million. Death and DALY rates have declined by 9.2% (95%CI 5.3–12.7) and 11.0% (95%CI 7.7–14.0) from 1990 to 2021, respectively. However, the global death rate increased to 7,062 per 100,000 population, with DALY rates rising to 125,042 per 100,000. A strong correlation exists between socio-demographic index (SDI) and DALY rates. Regions with lower SDI levels, particularly low and low-middle SDI regions, showed higher DALY rates, reaching 189,563 and 165,080 per 100,000, respectively. The leading causes of death among the older included ischemic heart disease, stroke, COVID-19, and chronic obstructive pulmonary disease. The primary risk factors contributing to DALYs were high systolic blood pressure, air pollution, and tobacco use.

Conclusions The global disease burden among adults aged 70 and older has shown declines in death and DALY rates, yet remains a significant challenge exacerbated by population growth, especially in less developed regions.

†Qin-Fen Chen and Chao Ni co-first author.

*Correspondence:
Yili Wu
wuyili@wmu.edu.cn
Xiao-Dong Zhou
zhouxiaodong@wmu.edu.cn
Weihong Song
weihong@wmu.edu.cn
Full list of author information is available at the end of the article

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Implementing preventive strategies that target primary risk factors early and enhancing treatment of underlying conditions are crucial.

Key points

- This study assesses the global disease burden and risk factors for adults aged 70 and older across 204 countries and territories.
- Ischemic heart disease, stroke, and COVID-19 are the leading causes of death and disability in this age group.
- Regional disparities in disease burden are linked to socio-demographic index (SDI), with lower SDI regions showing higher disease rates.

Keywords Global disease burden, Aging, Older population, Causes of death, Risk factors

Why does this paper matter?

This paper presents a comprehensive and novel global analysis of the disease burden and associated risk factors for adults aged 70 and older across 204 countries and territories. While previous studies have focused on aging-related diseases, this study uniquely incorporates a detailed, region-specific assessment, revealing new insights into the disparities in disease burden influenced by SDI. The study's novelty lies in its integration of recent Global Burden of Disease 2021 data, offering an updated and broader perspective on the health challenges faced by older adults. By examining the impact of key risk factors such as high blood pressure, air pollution, and tobacco use, the paper provides an in-depth exploration of their roles in shaping the disease burden across diverse geographic regions. Additionally, the analysis of the influence of SDI on disease rates and risk factors offers valuable insights into the socio-economic determinants of health among the aging population. These findings not only highlight critical global health disparities but also emphasize the need for region-specific interventions and policy adjustments to reduce the burden of chronic diseases among older adults.

Introduction

The global older population has been steadily increasing in recent decades and is projected to reach an impressive two billion by 2050 [1]. With this demographic shift, there has been a rapid rise in the occurrence of chronic diseases such as Alzheimer's disease, cardiovascular diseases, diabetes, cancer and other age-related disorders [2–4]. These trends pose substantial societal and economic challenges, such as soaring healthcare expenditures, a growing demand for long-term care services, and the pressing need for public health systems and policies that can adapt to the evolving needs of aging populations [5, 6].

Despite growing recognition of these challenges, the global epidemiological landscape concerning older adults remains inadequately defined, particularly for those aged 70 years and above [7, 8]. Most existing studies either focus on localized cohorts or aggregate data into broad age groups, thereby obscuring the distinct health patterns and risks faced by the oldest age segments [9–11]. This lack of disaggregated data hampers efforts to understand the true magnitude and nature of disease burden in this rapidly expanding population group [12]. The gap is especially pronounced in low- and middle-income countries, where surveillance systems are often underdeveloped, and older adults are underrepresented in national health data [13]. Importantly, the health status of older adults is not homogeneous and is strongly influenced by regional differences in socioeconomic development, healthcare access, environmental exposures, and lifestyle factors [14, 15]. In high-income regions, older individuals are more likely to benefit from preventive care, healthier diets, and greater access to medical services, which can reduce the prevalence or severity of age-related conditions. In contrast, older adults in resource-limited settings frequently face challenges such as food insecurity, pollution, and limited access to health services, which contribute to higher burdens of metabolic, respiratory, and infectious diseases. Moreover, the COVID-19 pandemic has further widened these health disparities, disproportionately affecting the oldest and most vulnerable populations worldwide [16].

Given these challenges, a nuanced understanding of the global burden of disease among adults aged 70 and older is essential for designing targeted interventions, allocating health resources effectively, and promoting healthy ageing. However, such assessments remain scarce and fragmented. Therefore, this study aims to conduct a detailed assessment of the global burden of disease in older adults, identify key regional patterns, and explore

associated risk factors across various demographic and geographic strata.

Methods

Data sources

The Global Burden of Disease (GBD) 2021 study is a comprehensive, multinational initiative aimed at quantifying the global burden of diseases, injuries, and risk factors. The GBD study integrated data from multiple sources, including national censuses, household surveys, vital statistics, disease registries, and environmental monitoring systems. These data were processed and analyzed using the GBD modeling framework, which employs sophisticated statistical techniques to estimate disease burden across regions and populations. This modeling framework incorporates a range of variables, such as demographic and epidemiological data, while accounting for uncertainties and gaps in available data. For confidentiality, the GBD study utilized the Data Input Sources Tool to ensure no personally identifiable information was used in the data collection process. As the study did not involve direct participant engagement or require the collection of personal data, it was exempt from standard ethics approval protocols and informed consent procedures. This methodology allows for the development of robust global health metrics while adhering to privacy standards and evolving ethical considerations in epidemiological research [17, 18].

Definitions in GBD 2021

The disease burden is measured by two primary metrics: deaths and disability-adjusted life years (DALYs). The Cause of Death Ensemble Model is utilized in the GBD study to estimate total deaths attributable to metabolic risk factors. It employs predictive factors to forecast deaths based on location, age, sex, and year. DALYs provide a comprehensive measure of the health burden caused by diseases. They are calculated by combining years of life lost (YLLs) and years lived with disability (YLDs). YLLs are determined by multiplying deaths from metabolic risk factors by life expectancy, while YLDs reflect the health impact at different severity levels [19].

The classification of causes of death and DALYs in the GBD study is structured into four hierarchical levels, each designed to be mutually exclusive and collectively exhaustive. At Level 1, the categorization includes three broad groupings: communicable, maternal, neonatal conditions, and nutritional diseases; non-communicable diseases; and injuries. Level 2 further breaks down these groupings into 22 specific disease and injury categories, such as cardiovascular diseases within the non-communicable diseases category. Moving to Level 3, the classification provides more detailed causes within the Level 2 categories, for instance, specifying conditions like stroke within the broader category of cardiovascular disease. Finally, Level 4 offers subcategories of Level 3 causes, such as identifying ischemic stroke within the general category of stroke. This hierarchical structure ensures thoroughness and specificity in categorizing the causes of death and disease burden, accommodating both broad categorizations and detailed analyses across global health metrics.

The GBD study categorizes risk factors associated with health outcomes into a detailed four-level hierarchy, supplemented by an aggregate encompassing all risk factors collectively [20]. This hierarchical structure systematically organizes and prioritizes risk factors based on their nature and global health impact. At Level 1, Risk Factors are divided into three main categories: environmental/occupational, behavioral, and metabolic risks. These categories encompass a wide array of factors influencing health outcomes worldwide. Moving to Level 2, these broad categories are further delineated into specific risk factors. For example, environmental/occupational risks may include factors like air pollution and non-optimal temperature. Behavioral risks encompass behaviors such as tobacco and dietary risks. Metabolic risks include high body mass index, high fasting plasma glucose, high low-density lipoprotein cholesterol, high systolic blood pressure and kidney dysfunction. Levels 3 and 4 within the hierarchy provide even greater specificity, detailing subcategories and specific risk factors within each Level 2 category. This structured approach allows for a comprehensive assessment of global health risk factors, facilitating targeted interventions and policies to mitigate their impact. The GBD Study groups countries into seven super regions based on geographic proximity and similarities in sociodemographic and epidemiological profiles: Central Europe, Eastern Europe, and Central Asia; High-income regions; Latin America and the Caribbean; North Africa and the Middle East; South Asia; Southeast Asia, East Asia, and Oceania; and Sub-Saharan Africa. This classification allows for more meaningful regional comparisons of disease burden and associated risk factors.

The socio-demographic index (SDI) is a comprehensive metric used to assess the development status of regions globally. It is calculated by averaging rankings across three key dimensions: the total fertility rate among women under 25 years old, educational attainment among individuals aged 15 and older, and per capita income adjusted for temporal changes. The data for SDI are sourced from the GBD Study 2021 and are updated every two years to reflect the latest available information. To ensure accuracy, adjustments are made to income, education, and fertility indicators to account for regional

variations and changes over time. SDI values are normalized on a scale from zero to one, where zero represents the least developed regions and one represents the most developed areas. This normalization allows countries and territories to be categorized into five quintiles based on their SDI score, facilitating comparative analyses of developmental levels across regions. In our statistical models, SDI serves as a key covariate to explore its relationship with disease burden, allowing us to assess how socio-economic factors influence health outcomes. This approach helps to elucidate socio-economic disparities and developmental progress on a global scale.

Statistical Analysis

Our study utilizes age-standardized mortality rates, DALYs rates, and percent changes, supplemented by 95% uncertainty intervals (UIs), for a comprehensive analysis. The Bayesian approach was selected over frequentist confidence intervals because it allows for the integration of prior knowledge and provides a more flexible framework to account for uncertainty in the data. The 95% UIs are derived from the 25 th and 75 th percentiles of the posterior distribution's ordered 1,000 sample values, from the posterior distribution, as described in existing literature, ensuring that our estimates reflect a higher level of precision, especially when the data are uncertain. A non-overlapping UI with zero indicates statistical significance, providing strong evidence of a meaningful association between the variables. We also calculated the age-standardized rates and corresponding 95% confidence intervals (CIs) based on the world standard population reported in the GBD Study 2021 for comparison across regions, and estimated average annual percentage changes to measure temporal trends. Population-attributable fractions were used to quantify the proportion of age-standardized DALYs that could be prevented with reduced exposure to metabolic risk factors. Additionally, we explored the relationship between the SDI and disease burden attributable to metabolic risk factors across different locations and years. Mapping figures were created using mapchart.net.

Results

Global and regional disease burden of adults aged 70 and older

In 2021, the global population of the older adults aged 70 and older reached 494.4 million (95%UI 482.1—508.2) people, marking a notable increase of 147.9% from 1990 to 2021. The rates of death and DALYs decreased by 9.2% (95% CI 5.3—12.7) and 11.0% (95% CI 7.7—14.0), respectively. The global death and DALYs numbers among adults aged 70 and older have shown a significant increase from 1990 to 2021, more than doubling in absolute numbers (Tables 1 and 2). Although the rates of both death and DALYs had been decreasing by 9.2% (95% CI 5.3—12.7) and 11.0% (95% CI 7.7—14.0), respectively during this period, a reversal of this trend occurred with the onset of COVID-19 (Supplement Fig. 1). From 2019 to 2021, there was an observable rise in the rate of the death burden. Regional disparities in death and DALY rates among adults aged 70 and older across GBD super-regions are depicted in Fig. 1. Notably, Sub-Saharan Africa exhibited substantially higher DALY rates, approximately two times greater than those in high-income regions, with rates of 195,412 per 100,000 (95% UI 182,720—208,818) compared to 93,629 per 100,000 (95% UI 87,021—101,389), respectively. From 1990 to 2021, changes in DALYs varied significantly, ranging from a decrease of 23.7% (95% CI 14.9—31.5) in Southeast Asia, East Asia, and Oceania to an increase of 11.7% (95% CI 6.1—17.8) in Sub-Saharan Africa. Analysis by SDI regions indicated a strong correlation between SDI and DALYs (Supplement Fig. 2). Regions with low SDI and low-middle SDI levels reported higher DALY rates compared to other SDI categories, reaching 189,563 (95% UI 177,647—201,889) per 100,000 and 165,080 (95% UI 155,948—174,752) per 100,000, respectively. The geographical disparities in death and DALY rates among individuals aged 70 and older, spanning 204 countries and territories, were illustrated in Fig. 2.

Leading causes of disease burden among adults aged 70 and older

In 2021, the leading causes of disease burden among adults aged 70 and older varied significantly. Cardiovascular diseases, respiratory infections including tuberculosis, neoplasms, chronic respiratory diseases, and neurologic disorders were identified as primary causes of death at a higher level. Specifically, ischemic heart disease, stroke, COVID-19, chronic obstructive pulmonary disease, and Alzheimer's disease and other dementias were among the top contributors. Over the span from 1990 to 2021, there were observable trends in the global burden associated with these causes, as illustrated in Fig. 3. Regional disparities of the top five causes among males and females exist (Fig. 4). Ischemic heart disease, stroke, and COVID-19 consistently ranked as the top three causes of DALYs, though their specific rankings varied between males and females across different regions. For instance, stroke was the leading cause of DALYs in Southeast Asian, East Asia and Oceania, whereas COVID-19 held the top position in South Asia and Sub-Saharan Africa. In South Asia, COVID-19 ranked highest among males, whereas chronic obstructive pulmonary disease (COPD) held the top position among females. Notably, Alzheimer's disease and other

Table 1 DALY counts and rates in 1990 and 2021 and the corresponding percentage change between 1990 and 2021 by location among adults aged ≥ 70

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Global	283,824,931 (268,819,750—299,623,856)	618,170,247 (579,871,381—659,986,135)	117.8% (110.4%—125.8%)	140,503 (133,075—148,324)	125,042 (117,295—133,501)	−11.0% (−14.0%—−7.7%)
Sex						
Female	152,271,920 (143,307,879—162,314,396)	310,065,645 (287,824,530—333,638,424)	103.6% (95.9%—112.2%)	127,650 (120,135—136,068)	111,959 (103,928—120,471)	−12.3% (−15.6%—−8.6%)
Male	131,553,010 (125,428,669—138,032,638)	308,104,602 (289,872,649—327,924,197)	134.2% (122.8%—147.0%)	159,038 (151,634—166,871)	141,707 (133,322—150,823)	−10.9% (−15.2%—−6.0%)
SDI Quintile						
Low SDI	17,470,563 (16,603,582—18,398,357)	41,578,480 (38,964,991—44,282,205)	138.0% (127.1%—149.3%)	187,209 (177,919—197,151)	189,563 (177,647—201,889)	1.3% (−3.4%—6.0%)
Low-middle SDI	42,878,052 (40,919,948—44,970,203)	115,713,032 (109,312,062—122,492,522)	169.9% (159.5%—180.4%)	163,505 (156,038—171,483)	165,080 (155,948—174,752)	1.0% (−2.9%—4.9%)
Middle SDI	67,772,482 (63,589,635—72,075,802)	181,762,641 (168,284,970—195,174,821)	168.2% (151.1%—187.8%)	148,315 (139,161—157,732)	128,915 (119,356—138,428)	−13.1% (−18.6%—−6.7%)
High-middle SDI	73,518,538 (69,444,973—77,831,945)	142,454,438 (132,694,761—153,127,755)	93.8% (84.4%—104.7%)	142,859 (134,944—151,241)	121,339 (113,026—130,431)	−15.1% (−19.2%—−10.3%)
High SDI	81,842,700 (77,285,542—87,102,111)	136,042,196 (126,328,532—147,398,846)	66.2% (63.1%—69.3%)	118,515 (111,916—126,131)	94,816 (88,046—102,731)	−20.0% (−21.5%—−18.5%)
Central Europe, Eastern Europe, and Central Asia	37,496,777 (35,797,001—39,477,564)	59,186,734 (56,004,534—62,525,584)	57.8% (54.6%—61.4%)	148,195 (141,477—156,024)	149,444 (141,409—157,874)	0.8% (−1.2%—3.1%)
Central Asia	3,085,370 (2,937,295—3,257,798)	5,282,716 (4,925,887—5,674,570)	71.2% (62.1%—80.5%)	136,929 (130,357—144,581)	155,414 (144,916—166,942)	13.5% (7.4%—19.6%)
Armenia	164,375 (156,776—173,421)	324,198 (299,837—350,036)	97.2% (86.3%—108.9%)	137,596 (131,234—145,168)	132,611 (122,647—143,180)	−3.6% (−9.0%—2.1%)
Azerbaijan	296,202 (281,817—312,892)	643,624 (592,697—694,451)	117.3% (104.6%—130.9%)	131,855 (125,452—139,285)	167,967 (154,677—181,231)	27.4% (19.9%—35.3%)
Georgia	461,393 (440,770—484,959)	623,388 (596,551—653,194)	35.1% (32.8%—37.2%)	139,769 (133,522—146,908)	167,126 (159,931—175,116)	19.6% (17.5%—21.4%)
Kazakhstan	867,316 (817,513—919,523)	1,410,391 (1,320,708—1,514,105)	62.6% (53.3%—73.3%)	144,559 (136,259—153,261)	173,416 (162,389—186,169)	20.0% (13.1%—27.9%)
Kyrgyzstan	201,539 (187,678—215,057)	253,983 (226,742—284,378)	26.0% (13.6%—40.1%)	139,366 (129,781—148,713)	132,324 (118,132—148,159)	−5.1% (−14.4%—5.6%)
Mongolia	81,557 (76,419—86,992)	124,250 (116,766—132,643)	52.3% (42.7%—62.6%)	158,709 (148,710—169,285)	147,676 (138,782—157,652)	−7.0% (−12.8%—−0.7%)
Tajikistan	186,052 (173,478—199,090)	312,655 (279,464—341,842)	68.0% (49.4%—86.7%)	145,273 (135,454—155,453)	157,154 (140,471—171,825)	8.2% (−3.8%—20.2%)
Turkmenistan	120,852 (115,289—126,803)	263,608 (231,678—300,957)	118.1% (89.7%—150.6%)	142,742 (136,171—149,770)	168,282 (147,899—192,125)	17.9% (2.5%—35.5%)
Uzbekistan	706,085 (670,661—747,415)	1,326,619 (1,184,769—1,482,909)	87.9% (70.7%—107.1%)	123,799 (117,588—131,045)	139,131 (124,254—155,522)	12.4% (2.1%—23.9%)
Central Europe	12,451,893 (11,910,991—13,081,923)	20,341,579 (19,339,224—21,516,114)	63.4% (61.7%—65.2%)	157,636 (150,789—165,612)	137,031 (130,279—144,944)	−13.1% (−14.0%—−12.1%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Albania	139,350 (132,422—146,468)	356,161 (318,879—392,221)	155.6% (132.5%—177.9%)	139,595 (132,654—146,726)	131,332 (117,585—144,630)	−5.9% (−14.4%—2.3%)
Bosnia and Herzegovina	229,066 (217,934—242,661)	542,145 (479,451—604,878)	136.7% (112.9%—161.0%)	136,624 (129,985—144,733)	139,516 (123,382—155,660)	2.1% (−8.2%—12.6%)
Bulgaria	1,100,086 (1,051,736—1,152,710)	1,898,790 (1,817,146—1,989,837)	72.6% (67.4%—78.1%)	166,805 (159,474—174,784)	184,453 (176,522—193,298)	10.6% (7.2%—14.1%)
Croatia	502,832 (482,922—526,331)	787,220 (739,973—841,057)	56.6% (52.0%—60.9%)	177,401 (170,377—185,691)	128,039 (120,355—136,796)	−27.8% (−29.9%—−25.8%)
Czechia	1,333,927 (1,275,465—1,399,206)	1,831,276 (1,718,870—1,959,490)	37.3% (34.1%—40.4%)	167,370 (160,035—175,561)	118,576 (111,298—126,878)	−29.2% (−30.8%—−27.6%)
Hungary	1,373,967 (1,309,950—1,444,472)	1,771,137 (1,679,569—1,882,510)	28.9% (27.1%—30.7%)	162,066 (154,514—170,382)	131,509 (124,710—139,779)	−18.9% (−20.0%—−17.7%)
North Macedonia	145,772 (139,258—153,256)	348,342 (318,148—382,923)	139.0% (117.9%—160.4%)	165,407 (158,015—173,899)	183,345 (167,452—201,546)	10.8% (1.1%—20.8%)
Montenegro	38,834 (36,581—41,270)	108,188 (102,579—114,383)	178.6% (164.9%—194.0%)	121,344 (114,302—128,954)	185,969 (176,328—196,618)	53.3% (45.7%—61.7%)
Poland	3,588,334 (3,415,707—3,781,578)	5,858,505 (5,547,231—6,242,598)	63.3% (61.4%—65.1%)	149,121 (141,947—157,152)	125,430 (118,765—133,653)	−15.9% (−16.8%—−14.9%)
Romania	2,219,518 (2,128,181—2,322,748)	3,681,236 (3,523,327—3,871,726)	65.9% (64.3%—67.5%)	155,542 (149,141—162,776)	145,421 (139,183—152,946)	−6.5% (−7.4%—−5.6%)
Serbia	886,224 (855,134—922,285)	1,737,837 (1,667,935—1,820,663)	96.1% (93.1%—98.8%)	178,436 (172,176—185,697)	160,736 (154,270—168,396)	−9.9% (−11.3%—−8.7%)
Slovakia	511,492 (488,982—537,617)	818,997 (778,861—865,648)	60.1% (57.7%—62.9%)	154,727 (147,918—162,630)	136,225 (129,549—143,984)	−12.0% (−13.3%—−10.4%)
Slovenia	183,362 (173,339—194,943)	305,667 (284,517—330,005)	66.7% (61.1%—72.7%)	133,896 (126,577—142,352)	101,727 (94,689—109,827)	−24.0% (−26.6%—−21.3%)
Eastern Europe	21,959,513 (20,957,543—23,136,928)	33,562,439 (31,574,395—35,522,971)	52.8% (48.0%—58.3%)	144,948 (138,335—152,720)	157,119 (147,812—166,297)	8.4% (5.0%—12.3%)
Belarus	963,802 (917,603—1,013,230)	1,544,447 (1,379,580—1,731,253)	60.2% (43.3%—78.9%)	138,535 (131,894—145,639)	161,864 (144,585—181,442)	16.8% (4.5%—30.4%)
Estonia	178,099 (169,903—187,443)	223,208 (209,293—237,642)	25.3% (21.2%—29.6%)	150,100 (143,193—157,975)	116,652 (109,380—124,196)	−22.3% (−24.8%—−19.6%)
Latvia	311,735 (296,884—327,822)	383,720 (362,519—406,499)	23.1% (20.2%—25.8%)	150,250 (143,092—158,003)	136,273 (128,744—144,363)	−9.3% (−11.5%—−7.3%)
Lithuania	349,313 (331,595—368,671)	530,193 (501,361—561,435)	51.8% (48.3%—55.2%)	137,631 (130,650—145,258)	132,007 (124,828—139,786)	−4.1% (−6.3%—−2.0%)
Republic of Moldova	323,488 (310,043—339,266)	483,708 (449,302—520,057)	49.5% (42.2%—57.3%)	154,008 (147,607—161,520)	138,804 (128,931—149,235)	−9.9% (−14.3%—−5.2%)
Russian Federation	13,921,416 (13,292,642—14,660,802)	22,876,767 (21,900,061—24,010,846)	64.3% (63.4%—65.3%)	144,177 (137,665—151,834)	159,896 (153,070—167,823)	10.9% (10.3%—11.5%)
Ukraine	5,911,660 (5,644,429—6,222,017)	7,520,397 (6,380,515—8,657,975)	27.2% (9.9%—46.0%)	147,482 (140,815—155,224)	154,212 (130,838—177,539)	4.6% (−9.7%—20.0%)
High-income	88,161,645 (83,197,014—94,024,281)	143,425,906 (133,303,567—155,313,372)	62.7% (59.9%—65.7%)	116,131 (109,591—123,853)	93,629 (87,021—101,389)	−19.4% (−20.8%—−17.9%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Australasia	1,675,377 (1,578,317—1,787,905)	2,953,286 (2,712,798—3,245,562)	76.3% (71.1%—81.4%)	114,959 (108,299—122,680)	81,059 (74,458—89,081)	−29.5% (−31.6%—−27.4%)
Australia	1,377,443 (1,297,160—1,471,210)	2,485,045 (2,278,875—2,731,940)	80.4% (74.9%—85.8%)	113,844 (107,209—121,594)	80,580 (73,895—88,586)	−29.2% (−31.4%—−27.1%)
New Zealand	297,933 (280,690—317,766)	468,241 (431,220—514,140)	57.2% (53.0%—62.3%)	120,411 (113,442—128,426)	83,695 (77,078—91,899)	−30.5% (−32.3%—−28.2%)
High-income Asia Pacific	11,596,806 (10,903,909—12,438,536)	26,801,808 (24,636,538—29,472,603)	131.1% (125.7%—137.0%)	103,030 (96,874—110,508)	76,710 (70,513—84,354)	−25.5% (−27.3%—−23.7%)
Brunei Darussalam	7,091 (6,583—7,635)	14,740 (13,412—16,210)	107.9% (89.0%—126.6%)	159,212 (147,811—171,437)	108,017 (98,285—118,796)	−32.2% (−38.3%—−26.0%)
Japan	9,694,460 (9,093,202—10,429,862)	21,999,370 (20,206,282—24,190,812)	126.9% (122.1%—131.9%)	98,150 (92,062—105,595)	76,117 (69,913—83,699)	−22.4% (−24.1%—−20.7%)
Republic of Korea	1,770,119 (1,686,501—1,867,405)	4,466,526 (4,102,507—4,895,792)	152.3% (143.1%—162.5%)	139,448 (132,861—147,112)	80,620 (74,050—88,369)	−42.2% (−44.3%—−39.9%)
Singapore	125,136 (119,017—132,757)	321,172 (293,446—355,606)	156.7% (146.3%—168.3%)	119,554 (113,707—126,835)	66,470 (60,731—73,596)	−44.4% (−46.7%—−41.9%)
High-income North America	26,193,631 (24,656,721—28,014,335)	45,669,683 (42,616,858—49,195,761)	74.4% (72.1%—76.4%)	112,797 (106,178—120,637)	105,452 (98,403—113,594)	−6.5% (−7.7%—−5.4%)
Canada	2,171,399 (2,044,056—2,317,334)	4,151,979 (3,840,314—4,534,102)	91.2% (87.1%—95.9%)	107,959 (101,627—115,214)	86,162 (79,695—94,092)	−20.2% (−21.9%—−18.2%)
United States of America	24,019,430 (22,612,329—25,692,541)	41,513,623 (38,773,125—44,687,333)	72.8% (70.5%—74.9%)	113,252 (106,617—121,140)	107,867 (100,746—116,113)	−4.8% (−6.0%—−3.6%)
Greenland	2,201 (2,081—2,322)	3,365 (3,070—4,010)	52.9% (40.8%—82.5%)	177,382 (167,697—187,106)	111,885 (102,080—133,351)	−36.9% (−41.9%—−24.7%)
Southern Latin America	3,293,599 (3,128,000—3,484,659)	6,160,062 (5,811,270—6,589,137)	87.0% (84.6%—89.9%)	124,987 (118,703—132,238)	112,054 (105,709—119,859)	−10.3% (−11.5%—−9.0%)
Argentina	2,319,033 (2,203,238—2,452,097)	4,160,111 (3,931,987—4,437,672)	79.4% (76.7%—82.1%)	125,404 (119,142—132,599)	116,736 (110,335—124,525)	−6.9% (−8.3%—−5.5%)
Chile	668,485 (633,346—708,171)	1,538,481 (1,435,982—1,659,358)	130.1% (126.0%—134.9%)	123,724 (117,221—131,070)	98,770 (92,189—106,530)	−20.2% (−21.6%—−18.5%)
Uruguay	305,924 (290,663—323,588)	461,130 (435,553—490,580)	50.7% (47.8%—53.6%)	124,631 (118,414—131,827)	122,717 (115,910—130,554)	−1.5% (−3.4%—0.4%)
Western Europe	45,402,232 (42,971,170—48,241,440)	61,841,068 (57,550,174—66,910,740)	36.2% (33.7%—38.8%)	121,573 (115,063—129,175)	93,988 (87,467—101,693)	−22.7% (−24.1%—−21.2%)
Andorra	3,075 (2,541—3,708)	8,130 (6,765—9,554)	164.4% (108.7%—229.0%)	96,015 (79,321—115,760)	87,118 (72,486—102,371)	−9.3% (−28.4%—12.9%)
Austria	966,845 (915,938—1,023,830)	1,199,952 (1,117,897—1,294,483)	24.1% (21.5%—26.8%)	126,817 (120,140—134,291)	96,465 (89,868—104,064)	−23.9% (−25.5%—−22.3%)
Belgium	1,208,086 (1,146,017—1,280,581)	1,504,621 (1,400,562—1,630,926)	24.5% (21.6%—27.6%)	125,723 (119,263—133,267)	93,578 (87,106—101,433)	−25.6% (−27.3%—−23.8%)
Cyprus	77,531 (73,959—81,490)	126,195 (114,268—138,813)	62.8% (50.8%—75.6%)	150,339 (143,413—158,016)	93,213 (84,403—102,533)	−38.0% (−42.5%—−33.1%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Denmark	707,939 (672,679—749,299)	788,577 (737,670—850,446)	11.4% (9.0%—13.9%)	126,645 (120,337—134,044)	91,729 (85,807—98,925)	−27.6% (−29.1%—−26.0%)
Finland	571,446 (541,184—607,266)	822,366 (760,838—895,515)	43.9% (40.0%—48.1%)	124,946 (118,329—132,778)	87,779 (81,211—95,586)	−29.7% (−31.6%—−27.7%)
France	5,933,665 (5,575,555—6,351,442)	8,663,238 (8,004,966—9,452,594)	46.0% (43.2%—48.9%)	114,038 (107,155—122,067)	87,221 (80,594—95,168)	−23.5% (−25.0%—−22.0%)
Germany	10,590,349 (10,066,783—11,206,592)	13,565,950 (12,650,385—14,608,652)	28.1% (25.5%—30.7%)	130,998 (124,522—138,621)	100,077 (93,323—107,769)	−23.6% (−25.2%—−22.0%)
Greece	1,142,912 (1,084,989—1,214,369)	1,853,156 (1,739,052—1,983,302)	62.1% (59.5%—64.7%)	120,647 (114,532—128,190)	107,040 (100,449—114,557)	−11.3% (−12.7%—−9.9%)
Iceland	20,434 (19,115—21,779)	30,786 (28,206—33,812)	50.7% (43.1%—58.6%)	111,840 (104,617—119,199)	81,373 (74,554—89,371)	−27.2% (−30.9%—−23.4%)
Ireland	370,325 (354,073—390,138)	441,745 (410,186—479,573)	19.3% (15.3%—23.6%)	137,818 (131,769—145,191)	84,975 (78,905—92,252)	−38.3% (−40.4%—−36.1%)
Israel	341,665 (323,360—363,244)	683,100 (632,391—745,287)	99.9% (93.9%—106.0%)	112,384 (106,363—119,482)	82,738 (76,596—90,270)	−26.4% (−28.6%—−24.2%)
Italy	6,448,195 (6,069,522—6,877,896)	9,871,703 (9,164,719—10,672,126)	53.1% (50.7%—55.6%)	116,412 (109,576—124,170)	93,868 (87,146—101,479)	−19.4% (−20.6%—−18.0%)
Luxembourg	43,375 (41,227—46,192)	59,375 (54,879—64,904)	36.9% (30.5%—43.1%)	127,015 (120,724—135,262)	89,338 (82,573—97,657)	−29.7% (−32.9%—−26.5%)
Malta	31,323 (29,691—33,268)	58,956 (54,099—64,714)	88.2% (78.6%—98.7%)	126,162 (119,588—133,997)	82,121 (75,356—90,142)	−34.9% (−38.2%—−31.3%)
Netherlands	1,528,505 (1,449,124—1,622,160)	2,338,471 (2,184,870—2,528,541)	53.0% (50.2%—56.1%)	119,152 (112,964—126,453)	95,216 (88,962—102,955)	−20.1% (−21.5%—−18.5%)
Norway	586,358 (554,158—623,860)	601,277 (554,957—653,941)	2.5% (0.0%—5.1%)	120,010 (113,419—127,685)	86,854 (80,163—94,462)	−27.6% (−29.4%—−25.8%)
Portugal	1,134,632 (1,080,657—1,201,013)	1,650,624 (1,540,329—1,786,763)	45.5% (42.0%—49.3%)	131,994 (125,715—139,716)	93,279 (87,046—100,972)	−29.3% (−31.0%—−27.5%)
Spain	3,866,150 (3,650,763—4,123,998)	6,021,614 (5,589,906—6,540,729)	55.8% (52.8%—59.0%)	112,171 (105,922—119,652)	88,989 (82,609—96,661)	−20.7% (−22.2%—−19.0%)
Sweden	1,232,139 (1,163,067—1,312,244)	1,365,259 (1,239,377—1,503,429)	10.8% (4.6%—16.8%)	112,881 (106,553—120,219)	85,606 (77,713—94,269)	−24.2% (−28.4%—−20.1%)
Switzerland	774,177 (729,230—828,071)	1,017,970 (935,934—1,111,259)	31.5% (28.0%—34.9%)	110,988 (104,544—118,714)	81,593 (75,017—89,070)	−26.5% (−28.4%—−24.6%)
United Kingdom	7,777,855 (7,394,734—8,226,908)	9,102,095 (8,503,787—9,801,615)	17.0% (15.0%—19.0%)	124,995 (118,838—132,212)	99,488 (92,948—107,133)	−20.4% (−21.8%—−19.0%)
Monaco	5,537 (4,600—6,264)	7,624 (6,925—8,453)	37.7% (20.4%—71.2%)	111,762 (92,848—126,435)	107,095 (97,279—118,742)	−4.2% (−16.2%—19.2%)
Latin America and Caribbean	13,143,589 (12,410,909—13,940,106)	43,346,795 (40,826,181—46,344,740)	229.8% (220.8%—240.4%)	120,554 (113,834—127,860)	125,627 (118,322—134,315)	4.2% (1.4%—7.6%)
Andean Latin America	1,174,806 (1,094,330—1,259,793)	4,971,945 (4,599,965—5,417,077)	323.2% (289.8%—361.4%)	115,126 (107,240—123,454)	151,569 (140,229—165,139)	31.7% (21.2%—43.5%)
Bolivia (Plurinational State of)	239,561 (221,075—259,573)	960,031 (872,512—1,086,113)	300.7% (259.4%—361.9%)	159,592 (147,276—172,923)	207,461 (188,549—234,707)	30.0% (16.6%—49.8%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Ecuador	300,486 (283,283—319,828)	1,147,183 (1,018,621—1,290,662)	281.8% (241.5%—327.3%)	110,958 (104,605—118,100)	122,707 (108,956—138,054)	10.6% (−1.1%—23.8%)
Peru	634,759 (577,036—699,692)	2,864,732 (2,597,887—3,145,005)	351.3% (302.1%—407.0%)	105,876 (96,248—116,706)	152,163 (137,989—167,050)	43.7% (28.0%—61.5%)
Caribbean	1,770,820 (1,674,175—1,870,154)	4,220,772 (3,834,337—4,644,990)	138.4% (120.3%—157.6%)	119,962 (113,415—126,691)	131,920 (119,842—145,179)	10.0% (1.6%—18.8%)
Antigua and Barbuda	4,596 (4,325—4,903)	7,318 (6,891—7,763)	59.2% (52.9%—66.6%)	126,337 (118,879—134,762)	130,286 (122,685—138,221)	3.1% (−1.0%—7.9%)
Bahamas	9,173 (8,536—9,854)	28,537 (25,343—32,146)	211.1% (176.5%—251.3%)	113,469 (105,590—121,888)	147,241 (130,759—165,863)	29.8% (15.3%—46.5%)
Barbados	25,738 (24,299—27,397)	36,991 (32,349—42,288)	43.7% (26.5%—62.3%)	120,433 (113,699—128,198)	115,429 (100,944—131,959)	−4.2% (−15.6%—8.2%)
Belize	5,337 (4,978—5,734)	16,506 (14,754—18,424)	209.3% (182.3%—236.7%)	101,303 (94,488—108,845)	118,225 (105,673—131,964)	16.7% (6.5%—27.1%)
Cuba	711,995 (674,386—754,098)	1,756,434 (1,621,471—1,907,974)	146.7% (130.9%—163.9%)	113,782 (107,772—120,510)	141,449 (130,580—153,653)	24.3% (16.4%—33.0%)
Dominica	5,566 (5,296—5,900)	7,109 (6,198—8,330)	27.7% (11.7%—49.8%)	144,341 (137,328—152,985)	157,540 (137,341—184,585)	9.1% (−4.5%—28.1%)
Dominican Republic	175,968 (159,435—195,486)	611,818 (544,658—680,148)	247.7% (210.2%—293.9%)	95,436 (86,470—106,021)	110,254 (98,151—122,568)	15.5% (3.1%—30.9%)
Grenada	6,814 (6,325—7,297)	10,086 (8,618—12,143)	48.0% (25.8%—80.1%)	136,967 (127,130—146,673)	170,622 (145,781—205,416)	24.6% (5.8%—51.6%)
Guyana	31,111 (28,869—33,571)	55,428 (42,085—72,032)	78.2% (33.9%—134.4%)	177,286 (164,512—191,304)	191,562 (145,450—248,947)	8.1% (−18.8%—42.2%)
Haiti	252,378 (228,199—275,893)	613,222 (476,227—788,785)	143.0% (87.0%—221.3%)	190,842 (172,558—208,624)	215,763 (167,561—277,534)	13.1% (−13.0%—49.5%)
Jamaica	124,532 (117,467—132,582)	233,378 (198,348—271,588)	87.4% (58.9%—117.5%)	107,047 (100,974—113,966)	128,315 (109,055—149,323)	19.9% (1.6%—39.1%)
Saint Lucia	7,755 (7,357—8,176)	18,646 (15,566—22,559)	140.4% (103.3%—191.2%)	148,371 (140,744—156,417)	137,272 (114,595—166,075)	−7.5% (−21.8%—12.1%)
Saint Vincent and the Grenadines	6,242 (5,909—6,636)	10,683 (9,782—11,754)	71.1% (57.3%—86.8%)	143,065 (135,421—152,080)	134,269 (122,944—147,732)	−6.1% (−13.7%—2.4%)
Suriname	15,508 (14,659—16,473)	42,867 (35,407—55,799)	176.4% (130.7%—258.6%)	126,962 (120,006—134,859)	128,837 (106,418—167,704)	1.5% (−15.3%—31.6%)
Trinidad and Tobago	67,266 (64,144—70,783)	154,307 (125,479—191,535)	129.4% (87.4%—184.3%)	142,490 (135,877—149,941)	137,905 (112,141—171,176)	−3.2% (−20.9%—20.0%)
Bermuda	4,759 (4,517—5,025)	9,091 (8,110—10,377)	91.0% (73.1%—114.8%)	136,992 (130,017—144,632)	100,903 (90,022—115,178)	−26.3% (−33.2%—−17.2%)
Puerto Rico	246,857 (232,440—262,664)	449,423 (396,692—512,687)	82.1% (63.7%—99.5%)	110,444 (103,994—117,516)	85,327 (75,315—97,338)	−22.7% (−30.5%—−15.4%)
Saint Kitts and Nevis	4,531 (4,311—4,776)	4,183 (3,802—4,595)	−7.7% (−16.5%—0.1%)	167,629 (159,501—176,697)	150,510 (136,802—165,363)	−10.2% (−18.8%—−2.6%)
United States Virgin Islands	4,731 (4,247—5,208)	11,911 (9,927—14,422)	151.8% (107.3%—212%)	118,084 (106,011—129,979)	92,364 (76,978—111,837)	−21.8% (−35.6%—−3.1%)
Central Latin America	4,792,417 (4,518,901—5,101,642)	17,573,375 (16,494,673—18,809,836)	266.7% (252.9%—281.4%)	118,632 (111,861—126,286)	128,637 (120,741—137,687)	8.4% (4.4%—12.8%)
Colombia	962,944 (910,296—1,025,363)	3,524,734 (3,167,302—3,926,664)	266.0% (235.0%—296.7%)	115,117 (108,823—122,579)	110,605 (99,389—123,218)	−3.9% (−12.1%—4.1%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Costa Rica	97,269 (90,704—104,710)	304,238 (283,751—329,386)	212.8% (203.8%—220.9%)	102,689 (95,757—110,544)	97,014 (90,482—105,034)	−5.5% (−8.2%—−3.1%)
El Salvador	165,239 (154,808—176,761)	462,965 (415,525—518,417)	180.2% (152.7%—209.5%)	104,289 (97,706—111,562)	120,902 (108,513—135,383)	15.9% (4.6%—28.1%)
Guatemala	229,458 (219,728—240,696)	850,451 (780,756—929,474)	270.6% (245.2%—298.2%)	156,389 (149,758—164,048)	144,275 (132,451—157,680)	−7.7% (−14.1%—−0.9%)
Honduras	107,291 (97,685—116,450)	572,354 (527,990—633,526)	433.5% (382.1%—495.2%)	109,959 (100,115—119,346)	181,620 (167,543—201,032)	65.2% (49.3%—84.3%)
Mexico	2,537,939 (2,395,252—2,704,672)	8,999,057 (8,540,882—9,527,707)	254.6% (251.9%—257.5%)	122,205 (115,335—130,234)	133,186 (126,405—141,010)	9.0% (8.2%—9.9%)
Nicaragua	65,502 (60,490—71,356)	313,320 (286,020—345,858)	378.3% (344.4%—419.0%)	89,001 (82,191—96,955)	125,487 (114,553—138,519)	41.0% (31.0%—53.0%)
Panama	77,922 (72,631—84,040)	246,746 (211,540—280,993)	216.7% (177.6%—253.9%)	96,562 (90,005—104,143)	94,547 (81,057—107,670)	−2.1% (−14.2%—9.4%)
Venezuela (Bolivarian Republic of)	548,852 (519,991—584,035)	2,299,510 (2,006,486—2,614,720)	319.0% (269.2%—371.6%)	115,617 (109,537—123,028)	143,188 (124,942—162,816)	23.8% (9.1%—39.4%)
Tropical Latin America	5,405,547 (5,105,468—5,751,950)	16,580,703 (15,594,200—17,794,669)	206.7% (203.9%—209.7%)	123,802 (116,930—131,736)	115,438 (108,569—123,889)	−6.8% (−7.6%—−5.9%)
Brazil	5,288,140 (4,993,917—5,624,023)	16,150,628 (15,191,537—17,300,930)	205.4% (202.6%—208.4%)	124,417 (117,495—132,320)	114,857 (108,037—123,038)	−7.7% (−8.5%—−6.8%)
Paraguay	117,407 (107,178—131,281)	430,074 (384,282—487,108)	266.3% (221.1%—315.0%)	101,250 (92,429—113,215)	142,460 (127,292—161,353)	40.7% (23.3%—59.4%)
Algeria	752,004 (680,099—829,540)	2,421,631 (2,146,310—2,689,675)	222.0% (181.8%—261.1%)	120,747 (109,201—133,197)	139,964 (124,051—155,456)	15.9% (1.4%—30.0%)
Bahrain	11,932 (11,369—12,490)	41,348 (37,513—45,355)	246.5% (216.6%—278.0%)	208,558 (198,716—218,314)	164,127 (148,903—180,031)	−21.3% (−28.1%—−14.2%)
Egypt	2,139,593 (2,077,572—2,208,188)	4,977,697 (4,450,290—5,642,715)	132.6% (108.8%—161.4%)	217,676 (211,366—224,655)	219,257 (196,026—248,550)	0.7% (−9.6%—13.2%)
Iran (Islamic Republic of)	1,209,729 (1,139,209—1,285,760)	5,123,765 (4,851,632—5,407,914)	323.5% (311.9%—336.9%)	124,675 (117,407—132,511)	140,313 (132,861—148,095)	12.5% (9.5%—16.1%)
Iraq	550,350 (497,689—611,746)	1,667,807 (1,444,523—1,866,345)	203.0% (160.3%—247.2%)	137,768 (124,585—153,137)	168,157 (145,644—188,175)	22.1% (4.8%—39.9%)
Jordan	65,392 (59,218—72,405)	368,767 (327,323—418,808)	463.9% (387.6%—544.4%)	133,329 (120,740—147,627)	120,077 (106,582—136,372)	−9.9% (−22.1%—2.9%)
Kuwait	20,618 (19,137—22,252)	97,018 (85,371—110,524)	370.5% (324.5%—422.9%)	103,257 (95,841—111,440)	98,264 (86,468—111,944)	−4.8% (−14.1%—5.8%)
Lebanon	139,562 (126,502—156,788)	558,094 (513,531—611,565)	299.9% (253.7%—346.5%)	145,932 (132,276—163,944)	145,134 (133,545—159,039)	−0.5% (−12.0%—11.1%)
Libya	100,894 (89,247—113,037)	331,551 (291,451—379,763)	228.6% (182.1%—282.9%)	116,094 (102,693—130,067)	151,294 (132,996—173,295)	30.3% (11.9%—51.8%)
Morocco	984,724 (917,245—1,064,424)	2,576,397 (2,267,393—2,794,185)	161.6% (130.2%—186.0%)	138,818 (129,306—150,054)	155,904 (137,205—169,082)	12.3% (−1.2%—22.8%)
Palestine	72,535 (62,310—83,266)	153,623 (139,660—168,515)	111.8% (83.4%—143.9%)	171,987 (147,744—197,433)	147,983 (134,532—162,328)	−14.0% (−25.5%—−0.9%)
Oman	44,461 (37,545—51,785)	106,877 (96,570—117,905)	140.4% (103.4%—191.4%)	154,059 (130,094—179,435)	154,790 (139,864—170,763)	0.5% (−15.0%—21.8%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Qatar	5,144 (4,604—5,640)	22,805 (19,623—26,310)	343.3% (273.4%—421.7%)	213,128 (190,735—233,669)	129,861 (111,740—149,818)	−39.1% (−48.7%—−28.3%)
Saudi Arabia	389,999 (332,245—445,767)	673,887 (610,604—746,127)	72.8% (50.8%—102.2%)	157,102 (133,837—179,566)	129,850 (117,656—143,770)	−17.3% (−27.9%—−3.3%)
Syrian Arab Republic	332,122 (291,083—376,846)	885,009 (760,676—1,041,469)	166.5% (119.1%—226.0%)	149,076 (130,656—169,152)	149,871 (128,816—176,367)	0.5% (−17.4%—23.0%)
Tunisia	287,895 (263,065—315,421)	1,017,000 (878,210—1,170,464)	253.3% (199.1%—319.4%)	119,087 (108,816—130,473)	146,068 (126,134—168,109)	22.7% (3.8%—45.6%)
Turkey	2,218,273 (2,053,848—2,376,526)	6,796,309 (6,042,158—7,556,554)	206.4% (174.2%—241.6%)	147,883 (136,921—158,433)	133,683 (118,849—148,637)	−9.6% (−19.1%—0.8%)
United Arab Emirates	19,683 (17,013—22,494)	84,723 (74,664—95,593)	330.4% (292.4%—368.1%)	161,745 (139,805—184,841)	175,493 (154,656—198,007)	8.5% (−1.1%—18.0%)
Yemen	325,497 (280,260—381,556)	1,087,916 (901,851—1,320,036)	234.2% (172.5%—316.5%)	166,851 (143,663—195,588)	182,889 (151,610—221,911)	9.6% (−10.6%—36.6%)
Afghanistan	658,067 (568,700—743,481)	1,036,755 (916,470—1,152,234)	57.5% (37.7%—84.3%)	211,752 (182,995—239,236)	256,278 (226,544—284,823)	21.0% (5.8%—41.6%)
Sudan	750,838 (671,319—843,915)	1,446,810 (1,229,066—1,703,727)	92.7% (62.1%—131.4%)	159,317 (142,444—179,066)	168,984 (143,552—198,992)	6.1% (−10.7%—27.4%)
North Africa and Middle East	11,085,374 (10,535,686—11,717,892)	31,505,176 (29,336,201—33,817,269)	184.2% (169.0%—199.1%)	153,502 (145,890—162,261)	154,938 (144,271—166,308)	0.9% (−4.5%—6.2%)
Bangladesh	3,642,001 (3,374,868—3,899,691)	9,669,250 (8,398,477—10,961,115)	165.5% (131.7%—201.6%)	160,066 (148,325—171,391)	131,204 (113,960—148,733)	−18.0% (−28.5%—−6.9%)
Bhutan	14,084 (12,434—15,913)	40,834 (36,521—44,526)	189.9% (151.8%—232.5%)	157,078 (138,667—177,471)	126,666 (113,288—138,119)	−19.4% (−30%—−7.5%)
India	30,532,051 (28,953,881—32,173,629)	96,515,275 (90,203,856—103,248,367)	216.1% (198.8%—233.7%)	169,512 (160,750—178,626)	161,982 (151,390—173,282)	−4.4% (−9.7%—0.9%)
Nepal	643,922 (572,631—727,504)	2,046,995 (1,848,719—2,311,220)	217.9% (176.6%—273.2%)	176,600 (157,048—199,523)	175,704 (158,685—198,384)	−0.5% (−13.4%—16.8%)
Pakistan	4,297,495 (4,029,195—4,575,125)	9,123,415 (8,183,549—10,243,010)	112.3% (91.0%—139.8%)	152,015 (142,524—161,835)	180,172 (161,611—202,282)	18.5% (6.6%—33.9%)
South Asia	39,129,554 (37,219,510—41,149,517)	117,395,770 (110,362,612—125,175,703)	200.0% (185.4%—214.6%)	166,596 (158,464—175,196)	160,345 (150,739—170,971)	−3.8% (−8.4%—0.9%)
Southeast Asia, East Asia, and Oceania	78,646,029 (72,976,000—84,517,486)	185,024,431 (167,664,667—203,967,961)	135.3% (111.1%—162.5%)	157,498 (146,143—169,256)	120,198 (108,920—132,504)	−23.7% (−31.5%—−14.9%)
East Asia	62,296,896 (56,855,549—67,540,692)	141,337,960 (125,503,738—157,240,690)	126.9% (99.3%—160.9%)	160,090 (146,107—173,565)	114,370 (101,557—127,238)	−28.6% (−37.2%—−17.8%)
China	60,459,838 (55,059,675—65,729,441)	136,942,698 (121,321,207—152,557,971)	126.5% (98.0%—161.4%)	161,099 (146,710—175,140)	114,785 (101,691—127,873)	−28.7% (−37.7%—−17.8%)
Democratic People's Republic of Korea	959,824 (829,850—1,084,246)	2,221,790 (1,972,338—2,492,544)	131.5% (99.8%—171.2%)	146,592 (126,741—165,595)	126,954 (112,700—142,425)	−13.4% (−25.3%—1.5%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Taiwan (Province of China)	877,234 (832,859—928,592)	2,173,473 (2,014,632—2,360,547)	147.8% (140.7%—154.6%)	120,283 (114,199—127,325)	86,061 (79,771—93,468)	−28.5% (−30.5%—−26.5%)
Oceania	177,764 (158,569—198,988)	511,940 (426,288—622,867)	188.0% (136.0%—257.7%)	170,993 (152,529—191,409)	185,581 (154,532—225,793)	8.5% (−11.1%—34.8%)
Viet Nam	2,903,255 (2,625,377—3,313,947)	6,000,122 (5,258,887—6,576,641)	106.7% (77.3%—131.0%)	142,385 (128,757—162,527)	134,300 (117,709—147,204)	−5.7% (−19.1%—5.4%)
Fiji	22,806 (20,920—24,998)	63,541 (51,414—78,214)	178.6% (123.6%—252.3%)	168,646 (154,699—184,857)	197,495 (159,803—243,100)	17.1% (−6.0%—48.1%)
Kiribati	2,852 (2,584—3,074)	4,475 (4,090—4,992)	56.9% (41.0%—78.2%)	197,778 (179,186—213,157)	179,856 (164,365—200,602)	−9.1% (−18.3%—3.3%)
Marshall Islands	1,418 (1,333—1,502)	1,992 (1,735—2,299)	40.5% (21.5%—61.4%)	193,164 (181,619—204,555)	173,533 (151,132—200,257)	−10.2% (−22.3%—3.2%)
Micronesia (Federated States of)	4,420 (4,044—4,897)	4,171 (3,664—4,860)	−5.6% (−18.4%—11.0%)	188,163 (172,187—208,471)	165,128 (145,024—192,395)	−12.2% (−24.1%—3.2%)
Papua New Guinea	104,016 (88,497—121,152)	342,960 (271,191—433,116)	229.7% (153.2%—333.1%)	173,247 (147,398—201,789)	197,070 (155,830—248,875)	13.8% (−12.6%—49.4%)
Samoa	6,144 (5,711—6,581)	10,048 (9,248—11,111)	63.5% (52.7%—81.8%)	162,141 (150,726—173,679)	147,064 (135,365—162,632)	−9.3% (−15.3%—0.8%)
Solomon Islands	9,117 (8,256—10,352)	22,827 (20,228—26,574)	150.4% (119.2%—189.2%)	194,202 (175,877—220,523)	172,245 (152,638—200,523)	−11.3% (−22.4%—2.5%)
Tonga	3,421 (3,121—3,737)	5,834 (5,144—6,598)	70.6% (48.6%—96.3%)	131,703 (120,153—143,893)	133,170 (117,422—150,609)	1.1% (−11.9%—16.4%)
Vanuatu	4,118 (3,755—4,629)	11,971 (10,576—13,844)	190.7% (151.1%—240.4%)	178,909 (163,135—201,137)	179,814 (158,858—207,951)	0.5% (−13.2%—17.7%)
American Samoa	1,259 (1,194—1,328)	3,162 (2,888—3,521)	151.3% (132.3%—176.6%)	143,363 (135,960—151,283)	142,805 (130,415—158,985)	−0.4% (−7.9%—9.6%)
Cook Islands	928 (875—987)	1,792 (1,604—2,001)	93.0% (74.1%—112.7%)	157,821 (148,683—167,715)	115,051 (102,985—128,513)	−27.1% (−34.2%—−19.7%)
Guam	3,277 (3,095—3,489)	10,378 (9,277—11,530)	216.7% (189.6%—243.1%)	117,719 (111,161—125,319)	84,813 (75,815—94,223)	−28.0% (−34.1%—−21.9%)
Nauru	321 (287—359)	441 (386—504)	37.2% (18.6%—59.2%)	203,612 (181,769—227,714)	194,942 (170,850—222,862)	−4.3% (−17.3%—11.1%)
Niue	249 (227—271)	188 (170—204)	−24.4% (−32.3%—−16.8%)	169,524 (154,891—184,485)	158,401 (143,032—171,827)	−6.6% (−16.2%—2.9%)
Northern Mariana Islands	635 (576—718)	2,876 (2,456—3,207)	353.1% (273.9%—422.5%)	132,073 (119,881—149,504)	135,774 (115,965—151,443)	2.8% (−15.2%—18.5%)
Palau	799 (716—889)	1,379 (1,196—1,584)	72.8% (49%—101.4%)	167,088 (149,753—185,997)	151,288 (131,212—173,770)	−9.5% (−21.9%—5.6%)
San Marino	2,398 (2,185—2,647)	3,834 (3,227—4,543)	59.9% (35.3%—88.4%)	102,652 (93,555—113,325)	73,587 (61,941—87,196)	−28.3% (−39.3%—−15.6%)
Tokelau	126 (114—138)	127 (111—144)	0.8% (−10.7%—14.1%)	167,437 (151,565—183,723)	138,758 (121,157—156,837)	−17.1% (−26.6%—−6.2%)
Tuvalu	520 (483—556)	792 (719—883)	52.3% (40.4%—67.0%)	192,672 (179,099—205,973)	155,501 (141,061—173,243)	−19.3% (−25.6%—−11.5%)
Southeast Asia	16,171,368 (15,264,508—17,132,592)	43,174,530 (38,711,628—47,442,456)	167.0% (143.8%—192.7%)	148,132 (139,825—156,937)	143,543 (128,705—157,733)	−3.1% (−11.5%—6.2%)
Cambodia	351,375 (322,490—382,409)	916,843 (784,701—1,041,401)	160.9% (119.1%—204.5%)	188,417 (172,928—205,058)	167,595 (143,440—190,364)	−11.1% (−25.3%—3.8%)
Indonesia	5,881,858 (5,516,987—6,283,936)	15,859,409 (13,214,157—18,458,659)	169.6% (127.2%—215.9%)	154,458 (144,876—165,016)	164,416 (136,992—191,362)	6.4% (−10.3%—24.7%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Lao People's Democratic Republic	179,752 (158,658—202,016)	312,828 (263,062—359,240)	74.0% (43.3%—110.1%)	202,560 (178,791—227,650)	162,398 (136,563—186,492)	−19.8% (−34.0%—−3.2%)
Malaysia	602,778 (573,938—637,037)	1,992,972 (1,847,286—2,141,143)	230.6% (216.5%—249.8%)	138,523 (131,895—146,396)	140,937 (130,635—151,416)	1.7% (−2.6%—7.6%)
Maldives	4,621 (4,354—4,899)	14,117 (12,345—16,060)	205.5% (170.9%—242.8%)	154,021 (145,114—163,278)	103,794 (90,767—118,083)	−32.6% (−40.2%—−24.4%)
Myanmar	1,922,445 (1,684,838—2,174,419)	3,729,525 (3,151,875—4,393,510)	94.0% (56.7%—138.6%)	196,556 (172,262—222,318)	166,892 (141,043—196,605)	−15.1% (−31.4%—4.4%)
Philippines	1,633,820 (1,524,326—1,758,910)	6,020,990 (5,529,215—6,506,933)	268.5% (238%—300.1%)	126,070 (117,621—135,723)	166,443 (152,848—179,876)	32.0% (21.1%—43.3%)
Sri Lanka	713,227 (671,952—758,450)	1,760,298 (1,356,723—2,150,545)	146.8% (92.4%—200.9%)	144,629 (136,259—153,800)	115,695 (89,170—141,344)	−20.0% (−37.6%—−2.5%)
Thailand	1,887,797 (1,745,172—2,053,165)	6,302,633 (5,329,096—7,337,166)	233.9% (183.7%—287.2%)	123,500 (114,169—134,318)	101,164 (85,537—117,769)	−18.1% (−30.4%—−5.0%)
Timor-Leste	13,654 (12,311—15,314)	72,064 (59,968—85,223)	427.8% (330.3%—541.8%)	160,026 (144,286—179,477)	159,365 (132,615—188,463)	−0.4% (−18.8%—21.1%)
Mauritius	48,632 (46,377—51,213)	124,795 (113,587—135,887)	156.6% (136.3%—174.1%)	147,679 (140,832—155,517)	124,447 (113,271—135,509)	−15.7% (−22.4%—−10.0%)
Seychelles	4,761 (4,498—5,020)	7,716 (7,179—8,338)	62.1% (52.7%—71.9%)	139,146 (131,481—146,745)	139,721 (130,001—150,984)	0.4% (−5.4%—6.5%)
Sub-Saharan Africa	16,161,962 (15,246,626—17,088,846)	38,285,436 (35,798,928—40,912,072)	136.9% (124.9%—149.7%)	174,873 (164,969—184,902)	195,412 (182,720—208,818)	11.7% (6.1%—17.8%)
Central Sub-Saharan Africa	1,573,132 (1,417,878—1,734,210)	4,033,159 (3,545,257—4,506,439)	156.4% (122.9%—195.4%)	194,070 (174,917—213,941)	213,918 (188,040—239,021)	10.2% (−4.2%—27.0%)
Angola	267,709 (235,720—301,400)	855,996 (737,413—977,202)	219.7% (167.6%—278.9%)	208,745 (183,802—235,015)	213,312 (183,761—243,516)	2.2% (−14.5%—21.1%)
Central African Republic	88,024 (81,712—93,315)	174,640 (151,983—200,782)	98.4% (71.1%—130.2%)	236,914 (219,924—251,155)	273,714 (238,203—314,687)	15.5% (−0.4%—34.0%)
Congo	93,526 (86,081—100,965)	204,741 (182,993—226,125)	118.9% (94.5%—145.5%)	214,497 (197,424—231,558)	212,889 (190,275—235,123)	−0.7% (−11.8%—11.3%)
Democratic Republic of the Congo	1,053,577 (939,173—1,172,397)	2,675,301 (2,294,715—3,019,877)	153.9% (114.6%—200.5%)	186,187 (165,969—207,184)	211,579 (181,480—238,831)	13.6% (−4.0%—34.5%)
Equatorial Guinea	16,662 (14,530—18,520)	38,858 (31,599—46,999)	133.2% (90.7%—186.0%)	222,052 (193,645—246,807)	202,968 (165,053—245,489)	−8.6% (−25.2%—12.1%)
Gabon	53,634 (48,107—58,580)	83,622 (74,512—92,716)	55.9% (35.5%—79.2%)	190,039 (170,454—207,565)	206,374 (183,891—228,817)	8.6% (−5.6%—24.8%)
Eastern Sub-Saharan Africa	6,171,012 (5,859,582—6,472,926)	14,109,010 (13,141,155—15,092,847)	128.6% (116.4%—141.7%)	198,140 (188,140—207,834)	204,769 (190,722—219,048)	3.3% (−2.2%—9.3%)
Burundi	254,950 (225,602—278,801)	349,610 (304,960—399,461)	37.1% (17.8%—61.9%)	225,327 (199,389—246,407)	194,073 (169,287—221,746)	−13.9% (−26%—1.7%)
Comoros	14,382 (12,447—16,269)	38,661 (34,160—44,567)	168.8% (126.0%—219.5%)	177,509 (153,629—200,790)	170,393 (150,554—196,419)	−4.0% (−19.3%—14.1%)
Djibouti	6,867 (5,907—8,060)	40,686 (35,053—48,155)	492.5% (393.8%—628.8%)	161,561 (138,978—189,637)	185,774 (160,055—219,879)	15.0% (−4.2%—41.4%)
Eritrea	63,524 (59,341—68,033)	194,563 (177,339—213,325)	206.3% (178.4%—236.6%)	225,564 (210,709—241,573)	211,937 (193,175—232,375)	−6.0% (−14.6%—3.2%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Ethiopia	1,860,664 (1,770,435—1,937,434)	3,693,787 (3,423,475—4,008,525)	98.5% (84.4%—117.2%)	240,753 (229,078—250,686)	187,352 (173,642—203,316)	−22.2% (−27.7%—−14.9%)
Kenya	557,601 (518,001—595,922)	1,834,519 (1,667,878—2,006,137)	229.0% (198.0%—264.7%)	150,485 (139,798—160,827)	196,605 (178,746—214,997)	30.6% (18.3%—44.8%)
Madagascar	408,988 (378,739—436,919)	748,417 (644,159—858,438)	83.0% (54.9%—110.4%)	190,519 (176,428—203,530)	205,761 (177,098—236,009)	8.0% (−8.6%—24.2%)
Malawi	292,651 (273,644—312,500)	758,142 (684,580—837,433)	159.1% (135.3%—184.3%)	188,165 (175,944—200,927)	243,705 (220,059—269,193)	29.5% (17.6%—42.1%)
Mozambique	452,601 (418,835—489,798)	1,059,949 (901,793—1,183,896)	134.2% (98.4%—164.7%)	186,231 (172,337—201,536)	246,568 (209,777—275,400)	32.4% (12.2%—49.6%)
Rwanda	250,805 (234,101—266,238)	461,832 (404,285—524,949)	84.1% (60.2%—112.1%)	227,136 (212,008—241,112)	186,907 (163,617—212,451)	−17.7% (−28.4%—−5.2%)
Somalia	163,328 (144,185—179,354)	561,088 (482,447—639,880)	243.5% (194.6%—299.6%)	232,735 (205,458—255,572)	294,506 (253,229—335,863)	26.5% (8.5%—47.2%)
United Republic of Tanzania	830,178 (765,119—903,384)	2,109,803 (1,912,139—2,398,828)	154.1% (126.3%—190.9%)	170,403 (157,049—185,430)	189,955 (172,159—215,978)	11.5% (−0.8%—27.6%)
Uganda	538,970 (486,535—595,417)	1,212,361 (1,035,324—1,388,382)	124.9% (91.4%—164.3%)	184,148 (166,233—203,434)	201,484 (172,062—230,737)	9.4% (−6.9%—28.6%)
Zambia	229,502 (210,698—248,531)	713,342 (604,259—811,880)	210.8% (161.4%—261.9%)	198,127 (181,893—214,554)	265,100 (224,561—301,720)	33.8% (12.5%—55.8%)
South Sudan	241,591 (209,245—280,374)	319,967 (274,942—370,156)	32.4% (10.8%—57.5%)	192,247 (166,508—223,109)	229,862 (197,517—265,918)	19.6% (0.0%—42.2%)
Southern Sub-Saharan Africa	1,589,008 (1,488,400—1,694,519)	5,577,945 (5,351,267—5,824,842)	251% (236.8%—268.2%)	122,405 (114,655—130,532)	209,155 (200,655—218,412)	70.9% (63.9%—79.2%)
Botswana	39,638 (33,263—44,915)	137,625 (120,495—153,025)	247.2% (192.1%—322.2%)	172,565 (144,812—195,542)	235,049 (205,794—261,351)	36.2% (14.6%—65.6%)
Lesotho	70,806 (63,828—79,521)	168,944 (146,443—189,491)	138.6% (101.7%—178.4%)	165,064 (148,796—185,380)	351,397 (304,595—394,133)	112.9% (80.0%—148.4%)
Namibia	45,780 (41,647—49,821)	147,030 (130,939—162,048)	221.2% (180.6%—265.4%)	172,402 (156,840—187,620)	245,598 (218,720—270,685)	42.5% (24.5%—62.1%)
South Africa	1,133,161 (1,062,030—1,213,020)	4,243,166 (4,084,125—4,439,952)	274.5% (260.6%—290.2%)	111,688 (104,677—119,559)	191,785 (184,596—200,679)	71.7% (65.4%—78.9%)
Zimbabwe	278,228 (242,161—305,326)	807,555 (718,848—894,663)	190.2% (151.2%—239.9%)	155,143 (135,032—170,253)	304,213 (270,796—337,027)	96.1% (69.7%—129.7%)
Western Sub-Saharan Africa	6,828,810 (6,339,040—7,292,986)	14,565,322 (13,538,684—15,727,030)	113.3% (96.5%—130.4%)	169,918 (157,731—181,468)	178,722 (166,125—192,976)	5.2% (−3.1%—13.6%)
Eswatini	21,396 (18,893—23,722)	73,624 (60,797—83,497)	244.1% (177.2%—311.1%)	180,967 (159,803—200,647)	327,469 (270,413—371,382)	81.0% (45.8%—116.2%)
Benin	180,660 (166,522—194,352)	403,575 (367,603—448,137)	123.4% (101.5%—148.6%)	176,112 (162,330—189,459)	180,323 (164,251—200,234)	2.4% (−7.6%—13.9%)
Burkina Faso	351,958 (323,408—384,011)	819,464 (732,309—920,968)	132.8% (105.3%—165.1%)	179,277 (164,734—195,604)	194,555 (173,863—218,654)	8.5% (−4.3%—23.6%)
Cameroon	312,910 (287,403—339,149)	1,023,941 (907,981—1,200,245)	227.2% (187.8%—284.6%)	169,967 (156,112—184,220)	204,667 (181,489—239,908)	20.4% (5.9%—41.5%)
Cabo Verde	17,385 (15,835—19,073)	29,960 (26,489—32,955)	72.3% (52.5%—93.7%)	116,267 (105,904—127,561)	146,265 (129,318—160,886)	25.8% (11.3%—41.4%)

Table 1 (continued)

	1990 DALY counts (95% UI)	2021 DALY counts (95% UI)	Annual percentage change of DALY counts (95% CI)	1990 DALY rates (95% UI)	2021 DALY rates (95% UI)	Annual percentage change of DALY rates (95% CI)
Chad	265,761 (244,793—288,842)	473,681 (414,741—547,194)	78.2% (54.6%—105.1%)	176,456 (162,534—191,781)	198,758 (174,027—229,604)	12.6% (−2.3%—29.6%)
Côte d'Ivoire	233,967 (211,985—256,396)	745,920 (668,516—856,784)	218.8% (182.0%—268.5%)	174,858 (158,429—191,620)	178,894 (160,331—205,483)	2.3% (−9.5%—18.2%)
Gambia	25,511 (22,515—28,970)	90,732 (76,816—102,094)	255.7% (195.6%—319.5%)	165,343 (145,928—187,767)	202,402 (171,357—227,748)	22.4% (1.7%—44.4%)
Ghana	423,081 (374,827—476,962)	1,263,257 (1,097,654—1,428,452)	198.6% (152.7%—248.4%)	166,958 (147,916—188,221)	183,504 (159,448—207,501)	9.9% (−7.0%—28.3%)
Guinea	287,907 (264,505—312,567)	536,656 (466,793—614,803)	86.4% (61.7%—114.7%)	173,631 (159,518—188,503)	206,790 (179,869—236,902)	19.1% (3.3%—37.2%)
Guinea-Bissau	36,769 (32,782—40,740)	60,545 (51,852—68,692)	64.7% (38.4%—88.7%)	224,812 (200,439—249,092)	238,506 (204,263—270,600)	6.1% (−10.8%—21.6%)
Liberia	100,997 (91,334—112,068)	149,667 (131,388—172,426)	48.2% (26.9%—75.2%)	169,748 (153,508—188,355)	187,671 (164,750—216,210)	10.6% (−5.4%—30.7%)
Mali	297,229 (274,490—321,208)	767,646 (686,321—860,069)	158.3% (129.9%—190.6%)	182,917 (168,924—197,674)	207,930 (185,902—232,965)	13.7% (1.2%—27.9%)
Mauritania	91,020 (81,801—101,781)	160,418 (138,995—187,599)	76.2% (49.2%—109.3%)	174,447 (156,777—195,071)	157,586 (136,541—184,287)	−9.7% (−23.5%—7.3%)
Niger	184,592 (170,996—200,101)	605,812 (532,204—693,767)	228.2% (188.1%—282.4%)	179,555 (166,331—194,641)	177,800 (156,196—203,614)	−1.0% (−13.1%—15.4%)
Nigeria	3,477,430 (3,129,801—3,852,602)	6,187,673 (5,682,236—6,813,267)	77.9% (58.3%—100.7%)	165,975 (149,383—183,882)	165,255 (151,757—181,963)	−0.4% (−11.4%—12.3%)
Sao Tome and Principe	4,925 (4,538—5,323)	7,448 (7,053—7,895)	51.2% (40.4%—61.9%)	149,065 (137,379—161,122)	162,467 (153,858—172,224)	9.0% (1.2%—16.7%)
Senegal	256,589 (236,496—278,113)	666,896 (584,128—763,135)	159.9% (124.7%—199.3%)	169,674 (156,387—183,907)	188,352 (164,976—215,532)	11.0% (−4.0%—27.8%)
Sierra Leone	193,653 (173,326—213,487)	306,239 (266,880—349,585)	58.1% (37.4%—85.7%)	179,255 (160,440—197,615)	178,599 (155,645—203,879)	−0.4% (−13.4%—17.0%)
Togo	86,244 (79,012—95,613)	265,623 (230,252—309,393)	208.0% (165.4%—262.9%)	170,118 (155,854—188,599)	183,786 (159,312—214,071)	8.0% (−6.9%—27.3%)

Abbreviations: CI confidence interval, *DALYs* deaths and disability-adjusted life years, *SDI* socio-demographic index, *UI* uncertainty intervals

dementias emerged as the leading cause among females in high-income regions.

Leading risk factors of disease burden among adults aged 70 and older

In 2021, risk factors contributed significantly to the disease burden in adults aged 70 and older, resulting in 300.6 million DALYs, which accounted for 60.8% of the total burden. These factors were categorized into environmental/occupational risks (111.9 million DALYs), behavioral risks (138.2 million DALYs), and metabolic risks (194.0 million DALYs). From 1990 to 2021, the top five risk factors contributing to DALYs consistently included high systolic blood pressure, air pollution, tobacco use, high fasting plasma glucose, and dietary risks (Fig. 5). During this period, high fasting plasma glucose ascended from the 5 th to the 4 th position. Similarly, high body-mass index rose from the 8 th to the 6 th position. There were significant variations between males and females in the distribution of these risk factors across different regions (Fig. 6). For males, high systolic blood pressure, tobacco use, and air pollution were identified as leading causes of DALYs, whereas for females, high systolic blood pressure, high fasting plasma glucose, and air pollution were predominant. Notably, air pollution remains one of the top two leading risk factors in South Asia, Southeast Asia, East Asia, Oceania, and Sub-Saharan Africa for both males and females. In high-income regions of Southeast Asia, East Asia, Oceania, and Sub-Saharan Africa, tobacco emerges as the primary risk factor among males and ranks second in South Asia. When examining global exposure to risk factors from 1990 to 2021, temporal trends reveal varied levels of exposure to both overall and specific risk factors (Supplement Fig. 3). The relationship between overall risk factor exposure and SDI trends is complex and multifaceted. Environmental

Table 2 Death counts and rates in 1990 and 2021 and the corresponding percentage change between 1990 and 2021 by location among adults aged ≥ 70

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Global	15,709,056(15,397,643—16,016,129)	34,912,199(33,748,055—36,184,288)	122.2% (113.5%—131.8%)	7,776(7,622—7,929)	7,062(6,827—7,319)	−9.2% (−12.7%—−5.3%)
Sex						
Female	8,455,795(8,239,734—8,692,020)	17,403,704(16,705,329—18,138,538)	105.8% (96.1%—116.4%)	7,089(6,907—7,287)	6,284(6,032—6,550)	−11.3% (−15.5%—−6.8%)
Male	7,253,261(7,042,615—7,483,068)	17,508,495(16,722,340—18,422,972)	141.4% (128.0%—156.1%)	8,769(8,514—9,046)	8,053(7,691—8,473)	−8.2% (−13.3%—−2.6%)
SDI Quintile						
Low SDI	923,824(893,414—959,990)	2,324,494(2,193,501—2,452,347)	151.6% (138.4%—165.4%)	9,899(9,574—10,287)	10,598(10,001—11,181)	7.1% (1.4%—12.9%)
Low-middle SDI	2,296,779(2,239,212—2,351,740)	6,445,367(6,186,265—6,729,249)	180.6% (168.4%—193.6%)	8,758(8,539—8,968)	9,195(8,826—9,600)	5.0% (0.4%—9.8%)
Middle SDI	3,673,084(3,507,878—3,843,713)	10,173,927(9,577,189—10,785,438)	177.0% (155.9%—200.7%)	8,038(7,677—8,412)	7,216(6,793—7,650)	−10.2% (−17.1%—−2.5%)
High-middle SDI	4,159,578(4,039,259—4,282,533)	8,279,340(7,851,291—8,708,974)	99.0% (87.5%—111.8%)	8,083(7,849—8,322)	7,052(6,688—7,418)	−12.8% (−17.8%—−7.2%)
High SDI	4,636,203(4,620,854—4,650,996)	7,652,807(7,594,645—7,712,108)	65.1% (63.7%—66.4%)	6,714(6,691—6,735)	5,334(5,293—5,375)	−20.6% (−21.2%—−19.9%)
Central Europe, Eastern Europe, and Central Asia	2,159,127(2,154,570—2,163,704)	3,545,586(3,458,072—3,638,555)	64.2% (60.2%—68.7%)	8,533(8,515—8,551)	8,952(8,731—9,187)	4.9% (2.3%—7.8%)
Central Asia	178,334(176,164—180,373)	312,215(292,209—331,694)	75.1% (64.2%—86.0%)	7,914(7,818—8,005)	9,185(8,597—9,758)	16.1% (8.8%—23.3%)
Armenia	9,602(9,468—9,753)	19,494(18,223—20,887)	103.0% (89.5%—117.5%)	8,038(7,926—8,164)	7,974(7,454—8,544)	−0.8% (−7.4%—6.3%)
Azerbaijan	17,276(16,777—17,836)	38,707(35,727—41,496)	124.0% (108.0%—140.1%)	7,691(7,468—7,940)	10,101(9,324—10,829)	31.3% (22.0%—40.8%)
Georgia	26,855(26,490—27,240)	38,938(38,360—39,544)	45.0% (42.2%—47.5%)	8,135(8,024—8,252)	10,439(10,284—10,602)	28.3% (25.9%—30.5%)
Kazakhstan	49,529(47,920—51,191)	82,839(77,864—87,557)	67.3% (56.2%—79.5%)	8,255(7,987—8,532)	10,186(9,574—10,766)	23.4% (15.3%—32.4%)
Kyrgyzstan	11,724(11,100—12,358)	14,792(12,948—16,597)	26.2% (11.3%—43.2%)	8,107(7,676—8,545)	7,707(6,746—8,647)	−4.9% (−16.1%—7.9%)
Mongolia	4,629(4,391—4,842)	7,113(6,708—7,517)	53.6% (42.1%—66.0%)	9,008(8,544—9,422)	8,454(7,973—8,934)	−6.2% (−13.2%—1.4%)
Tajikistan	11,113(10,511—11,685)	18,067(15,986—20,027)	62.6% (41.8%—83.2%)	8,677(8,207—9,124)	9,081(8,035—10,066)	4.7% (−8.7%—17.9%)
Turkmenistan	6,862(6,775—6,941)	15,998(13,612—18,543)	133.2% (98.6%—171.2%)	8,104(8,002—8,198)	10,213(8,689—11,837)	26.0% (7.3%—46.6%)
Uzbekistan	40,743(40,477—41,007)	76,267(67,443—85,886)	87.2% (66.1%—110.5%)	7,144(7,097—7,190)	7,999(7,073—9,007)	12.0% (−0.7%—25.9%)
Central Europe	723,695(721,249—726,199)	1,204,694(1,194,231—1,214,812)	66.5% (64.9%—68.0%)	9,162(9,131—9,193)	8,115(8,045—8,184)	−11.4% (−12.3%—−10.6%)
Albania	8,004(7,814—8,210)	21,443(19,182—23,556)	167.9% (139%—195.8%)	8,018(7,828—8,224)	7,907(7,073—8,686)	−1.4% (−12.0%—8.9%)
Bosnia and Herzegovina	12,620(12,437—12,817)	31,429(27,545—35,357)	149.0% (117.9%—180.2%)	7,527(7,418—7,644)	8,088(7,089—9,099)	7.4% (−6.0%—20.9%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Bulgaria	61,918(60,572—63,161)	116,828(114,234—119,751)	88.7% (82.5%—95.1%)	9,389(9,185—9,577)	11,349(11,097—11,633)	20.9% (16.9%—25.0%)
Croatia	30,794(30,428—31,177)	46,934(45,777—48,066)	52.4% (48.1%—56.5%)	10,864(10,735—10,999)	7,634(7,446—7,818)	−29.7% (−31.7%—−27.9%)
Czechia	77,655(77,012—78,344)	101,961(100,163—103,865)	31.3% (28.6%—34.0%)	9,743(9,663—9,830)	6,602(6,486—6,725)	−32.2% (−33.7%—−30.8%)
Hungary	79,045(78,434—79,628)	102,274(101,101—103,391)	29.4% (27.6%—31.1%)	9,324(9,252—9,392)	7,594(7,507—7,677)	−18.6% (−19.7%—−17.5%)
North Macedonia	8,500(8,304—8,709)	20,155(18,225—22,123)	137.1% (113.3%—160.5%)	9,645(9,422—9,883)	10,608(9,593—11,644)	10.0% (−1.0%—20.9%)
Montenegro	2,257(2,186—2,336)	6,767(6,487—7,061)	199.8% (184.7%—217.7%)	7,052(6,830—7,300)	11,632(11,150—12,138)	64.9% (56.6%—74.8%)
Poland	206,168(205,141—207,174)	344,399(342,260—346,481)	67.0% (65.6%—68.3%)	8,568(8,525—8,610)	7,374(7,328—7,418)	−13.9% (−14.7%—−13.3%)
Romania	129,524(128,771—130,244)	223,676(221,911—225,608)	72.7% (70.9%—74.5%)	9,077(9,024—9,127)	8,836(8,766—8,912)	−2.7% (−3.7%—−1.7%)
Serbia	55,677(55,212—56,182)	106,454(105,118—107,868)	91.2% (88.2%—94.0%)	11,210(11,117—11,312)	9,846(9,723—9,977)	−12.2% (−13.6%—−10.9%)
Slovakia	29,566(29,215—29,924)	47,280(46,570—47,961)	59.9% (57.1%—63.2%)	8,944(8,838—9,052)	7,864(7,746—7,977)	−12.1% (−13.6%—−10.3%)
Slovenia	10,395(10,193—10,621)	17,560(16,911—18,189)	68.9% (61.6%—75.9%)	7,591(7,443—7,756)	5,844(5,628—6,053)	−23.0% (−26.3%—−19.8%)
Eastern Europe	1,257,097(1,254,587—1,259,488)	2,028,677(1,951,764—2,110,856)	61.4% (55.3%—67.9%)	8,298(8,281—8,314)	9,497(9,137—9,882)	14.5% (10.1%—19.1%)
Belarus	57,083(56,538—57,665)	96,460(84,623—109,165)	69.0% (48.2%—91.4%)	8,205(8,127—8,289)	10,109(8,869—11,441)	23.2% (8.1%—39.5%)
Estonia	10,571(10,361—10,797)	13,498(13,079—13,991)	27.7% (22.9%—33.1%)	8,909(8,732—9,099)	7,054(6,835—7,312)	−20.8% (−23.8%—−17.4%)
Latvia	18,592(18,272—18,900)	23,457(22,953—23,967)	26.2% (22.9%—29.7%)	8,961(8,807—9,110)	8,330(8,151—8,512)	−7.0% (−9.5%—−4.5%)
Lithuania	20,970(20,670—21,283)	32,689(32,036—33,316)	55.9% (51.7%—59.9%)	8,262(8,144—8,385)	8,139(7,976—8,295)	−1.5% (−4.1%—1.0%)
Republic of Moldova	18,290(18,116—18,473)	27,739(26,119—29,386)	51.7% (42.9%—60.7%)	8,708(8,625—8,795)	7,960(7,495—8,433)	−8.6% (−13.9%—−3.2%)
Russian Federation	790,897(788,939—792,695)	1,375,129(1,371,944—1,378,102)	73.9% (73.3%—74.5%)	8,191(8,171—8,210)	9,611(9,589—9,632)	17.3% (16.9%—17.8%)
Ukraine	340,694(339,155—342,155)	459,706(384,498—539,799)	34.9% (12.6%—58.7%)	8,500(8,461—8,536)	9,427(7,884—11,069)	10.9% (−7.4%—30.4%)
High-income	4,995,736(4,990,438—5,001,772)	8,129,921(8,101,656—8,158,286)	62.7% (62.2%—63.3%)	6,581(6,574—6,589)	5,307(5,289—5,326)	−19.4% (−19.6%—−19.1%)
Australasia	91,750(91,312—92,155)	158,993(158,299—159,647)	73.3% (72.3%—74.4%)	6,296(6,266—6,323)	4,364(4,345—4,382)	−30.7% (−31.1%—−30.2%)
Australia	75,425(75,017—75,800)	133,502(132,895—134,055)	77.0% (75.8%—78.2%)	6,234(6,200—6,265)	4,329(4,309—4,347)	−30.6% (−31.0%—−30.1%)
New Zealand	16,325(16,175—16,477)	25,491(25,196—25,801)	56.1% (53.9%—58.5%)	6,598(6,537—6,659)	4,556(4,504—4,612)	−30.9% (−31.9%—−29.9%)
High-income Asia Pacific	630,794(629,131—632,380)	1,495,429(1,488,748—1,501,566)	137.1% (135.9%—138.2%)	5,604(5,589—5,618)	4,280(4,261—4,298)	−23.6% (−24.0%—−23.3%)
Brunei Darussalam	404(377—434)	747(679—815)	85.1% (65.8%—105.2%)	9,068(8,459—9,741)	5,477(4,977—5,972)	−39.6% (−45.9%—−33.0%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Japan	527,991(526,325—529,547)	1,242,791(1,236,743—1,248,009)	135.4% (134.2%—136.6%)	5,346(5,329—5,361)	4,300(4,279—4,318)	−19.6% (−19.9%—−19.1%)
Republic of Korea	95,731(95,388—96,118)	235,683(232,425—238,643)	146.2% (142.6%—149.6%)	7,542(7,515—7,572)	4,254(4,195—4,307)	−43.6% (−44.4%—−42.8%)
Singapore	6,668(6,607—6,731)	16,208(15,933—16,489)	143.1% (138.3%—147.9%)	6,370(6,312—6,431)	3,354(3,297—3,412)	−47.3% (−48.4%—−46.3%)
High-income North America	1,442,198(1,439,674—1,444,734)	2,499,200(2,478,639—2,519,270)	73.3% (71.9%—74.7%)	6,210(6,200—6,221)	5,771(5,723—5,817)	−7.1% (−7.8%—−6.3%)
Canada	119,972(119,490—120,418)	226,811(224,394—229,166)	89.1% (87.1%—91.1%)	5,965(5,941—5,987)	4,707(4,657—4,756)	−21.1% (−21.9%—−20.2%)
United States of America	1,322,073(1,319,524—1,324,679)	2,272,179(2,252,440—2,291,991)	71.9% (70.3%—73.4%)	6,234(6,222—6,246)	5,904(5,853—5,955)	−5.3% (−6.1%—−4.5%)
Greenland	120(115—126)	172(158—211)	43.0% (30.1%—76.5%)	9,672(9,261—10,153)	5,708(5,253—7,017)	−41.0% (−46.3%—−27.2%)
Southern Latin America	184,041(183,503—184,607)	356,733(352,278—361,392)	93.8% (91.2%—96.3%)	6,984(6,964—7,006)	6,489(6,408—6,574)	−7.1% (−8.4%—−5.9%)
Argentina	129,915(129,448—130,432)	243,093(239,330—247,172)	87.1% (83.9%—90.2%)	7,025(7,000—7,053)	6,821(6,716—6,936)	−2.9% (−4.6%—−1.3%)
Chile	36,671(36,464—36,856)	85,217(84,575—85,927)	132.4% (130.4%—134.3%)	6,787(6,749—6,821)	5,471(5,430—5,516)	−19.4% (−20.1%—−18.7%)
Uruguay	17,447(17,304—17,600)	28,403(27,826—28,998)	62.8% (59.1%—66.4%)	7,108(7,050—7,170)	7,559(7,405—7,717)	6.3% (3.9%—8.7%)
Western Europe	2,646,953(2,642,444—2,651,381)	3,619,566(3,605,086—3,634,483)	36.7% (36.1%—37.3%)	7,088(7,076—7,100)	5,501(5,479—5,524)	−22.4% (−22.7%—−22.1%)
Andorra	158(126—199)	457(367—548)	188.7% (112.7%—288.4%)	4,939(3,927—6,199)	4,893(3,932—5,869)	−0.9% (−27.0%—33.3%)
Austria	57,396(57,028—57,739)	70,308(69,505—71,149)	22.5% (20.9%—24.1%)	7,528(7,480—7,573)	5,652(5,588—5,720)	−24.9% (−25.9%—−24%)
Belgium	71,416(70,964—71,873)	86,414(85,546—87,282)	21.0% (19.5%—22.5%)	7,432(7,385—7,480)	5,374(5,320—5,428)	−27.7% (−28.6%—−26.8%)
Cyprus	4,576(4,478—4,687)	6,992(6,428—7,580)	52.8% (40.0%—66.5%)	8,874(8,683—9,088)	5,164(4,748—5,599)	−41.8% (−46.7%—−36.6%)
Denmark	41,096(40,745—41,429)	44,890(44,234—45,565)	9.2% (7.5%—11.1%)	7,352(7,289—7,411)	5,222(5,145—5,300)	−29.0% (−30.1%—−27.7%)
Finland	32,146(31,845—32,468)	45,423(44,666—46,171)	41.3% (38.6%—43.9%)	7,029(6,963—7,099)	4,848(4,768—4,928)	−31.0% (−32.3%—−29.7%)
France	354,556(353,412—355,840)	505,168(502,274—507,929)	42.5% (41.5%—43.4%)	6,814(6,792—6,839)	5,086(5,057—5,114)	−25.4% (−25.9%—−24.9%)
Germany	632,507(630,644—634,338)	793,929(789,852—797,691)	25.5% (24.8%—26.3%)	7,824(7,801—7,846)	5,857(5,827—5,885)	−25.1% (−25.6%—−24.7%)
Greece	66,253(65,756—66,746)	116,673(114,948—118,372)	76.1% (73.2%—78.9%)	6,994(6,941—7,046)	6,739(6,640—6,837)	−3.6% (−5.2%—−2.1%)
Iceland	1,191(1,153—1,228)	1,751(1,660—1,852)	47.0% (37.7%—57.1%)	6,517(6,312—6,721)	4,628(4,387—4,896)	−29.0% (−33.5%—−24.1%)
Ireland	21,296(21,087—21,510)	24,361(23,965—24,808)	14.4% (12.3%—16.7%)	7,926(7,848—8,005)	4,686(4,610—4,772)	−40.9% (−42.0%—−39.7%)
Israel	18,892(18,709—19,063)	37,946(37,224—38,722)	100.9% (96.8%—105.2%)	6,214(6,154—6,270)	4,596(4,509—4,690)	−26.0% (−27.5%—−24.4%)
Italy	364,000(362,877—365,283)	586,564(583,569—589,513)	61.1% (60.2%—62.1%)	6,571(6,551—6,595)	5,578(5,549—5,606)	−15.1% (−15.6%—−14.6%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Luxembourg	2,528(2,472—2,587)	3,442(3,286—3,601)	36.1% (29.2%—42.7%)	7,403(7,238—7,577)	5,179(4,944—5,419)	−30.0% (−33.6%—−26.7%)
Malta	1,782(1,726—1,835)	3,188(3,024—3,362)	78.9% (68.7%—89.7%)	7,176(6,952—7,391)	4,440(4,213—4,683)	−38.1% (−41.7%—−34.4%)
Netherlands	88,678(88,175—89,140)	135,374(134,103—136,799)	52.7% (50.9%—54.4%)	6,913(6,874—6,949)	5,512(5,460—5,570)	−20.3% (−21.2%—−19.4%)
Norway	33,494(33,206—33,833)	33,860(33,416—34,356)	1.1% (−0.4%—2.8%)	6,855(6,796—6,925)	4,891(4,827—4,963)	−28.7% (−29.7%—−27.4%)
Portugal	65,630(65,221—66,016)	98,287(97,227—99,332)	49.8% (47.9%—51.6%)	7,635(7,587—7,680)	5,554(5,494—5,613)	−27.3% (−28.1%—−26.3%)
Spain	220,618(219,684—221,612)	355,832(353,351—358,225)	61.3% (59.8%—62.7%)	6,401(6,374—6,430)	5,259(5,222—5,294)	−17.8% (−18.6%—−17.2%)
Sweden	71,014(70,564—71,497)	77,145(72,355—82,091)	8.6% (1.8%—15.8%)	6,506(6,465—6,550)	4,837(4,537—5,147)	−25.6% (−30.3%—−20.7%)
Switzerland	44,536(44,194—44,895)	57,016(56,279—57,799)	28.0% (26.2%—30.0%)	6,385(6,336—6,436)	4,570(4,511—4,633)	−28.4% (−29.5%—−27.3%)
United Kingdom	450,543(449,175—452,056)	530,686(528,135—533,315)	17.8% (17.1%—18.5%)	7,241(7,219—7,265)	5,801(5,773—5,829)	−19.9% (−20.3%—−19.4%)
Monaco	333(260—376)	465(418—515)	39.6% (19.3%—84.7%)	6,723(5,243—7,595)	6,534(5,870—7,231)	−2.8% (−16.9%—28.5%)
Latin America and Caribbean	708,091(703,298—712,986)	2,465,493(2,380,838—2,560,986)	248.2% (236.6%—262.2%)	6,495(6,451—6,540)	7,145(6,900—7,422)	10.0% (6.3%—14.5%)
Andean Latin America	65,707(62,279—69,614)	304,636(281,118—332,527)	363.6% (320.8%—411.4%)	6,439(6,103—6,822)	9,287(8,570—10,137)	44.2% (30.9%—59.1%)
Bolivia (Plurinational State of)	13,235(12,282—14,340)	56,784(51,305—65,164)	329.0% (279.7%—399.0%)	8,817(8,182—9,553)	12,271(11,087—14,082)	39.2% (23.2%—61.9%)
Ecuador	16,975(16,743—17,205)	66,933(58,307—75,963)	294.3% (243.9%—349.9%)	6,268(6,183—6,353)	7,159(6,237—8,125)	14.2% (−0.4%—30.3%)
Peru	35,496(32,136—38,989)	180,919(163,647—200,754)	409.7% (343.3%—483.4%)	5,921(5,360—6,503)	9,610(8,692—10,663)	62.3% (41.2%—85.8%)
Caribbean	98,881(97,003—100,626)	249,869(227,622—272,210)	152.7% (130.2%—175.9%)	6,699(6,571—6,817)	7,810(7,114—8,508)	16.6% (6.2%—27.3%)
Antigua and Barbuda	271(257—283)	410(400—422)	51.5% (43.6%—60.0%)	7,449(7,060—7,784)	7,308(7,122—7,509)	−1.9% (−7.0%—3.6%)
Bahamas	486(462—511)	1,689(1,480—1,921)	247.6% (201.8%—299.8%)	6,013(5,711—6,317)	8,717(7,635—9,909)	45.0% (25.9%—66.8%)
Barbados	1,500(1,456—1,547)	2,145(1,844—2,477)	43.0% (22.2%—65.5%)	7,019(6,814—7,238)	6,694(5,753—7,728)	−4.6% (−18.5%—10.4%)
Belize	294(285—304)	948(845—1,049)	222.8% (187.0%—258.7%)	5,575(5,402—5,778)	6,790(6,055—7,516)	21.8% (8.3%—35.4%)
Cuba	40,183(39,813—40,516)	109,090(100,507—117,817)	171.5% (150.6%—193.7%)	6,422(6,362—6,475)	8,785(8,094—9,488)	36.8% (26.3%—48%)
Dominica	323(311—332)	424(366—511)	31.1% (12.2%—57.9%)	8,381(8,062—8,622)	9,391(8,109—11,333)	12.1% (−4.1%—35.0%)
Dominican Republic	9,504(8,629—10,396)	34,538(30,729—38,757)	263.4% (214.1%—323.8%)	5,155(4,680—5,638)	6,224(5,538—6,984)	20.7% (4.4%—40.8%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Grenada	411(386—438)	571(477—711)	38.9% (14.5%—73.2%)	8,269(7,754—8,802)	9,664(8,063—12,026)	16.9% (−3.7%—45.7%)
Guyana	1,742(1,625—1,864)	3,224(2,301—4,357)	85% (30.7%—152.4%)	9,929(9,260—10,625)	11,142(7,953—15,059)	12.2% (−20.8%—53.1%)
Haiti	13,663(12,278—15,038)	35,143(26,166—46,529)	157.2% (90.3%—249.1%)	10,331(9,284—11,371)	12,365(9,207—16,371)	19.7% (−11.4%—62.4%)
Jamaica	7,012(6,857—7,178)	14,397(11,794—17,079)	105.3% (67.9%—143.8%)	6,028(5,894—6,170)	7,916(6,485—9,390)	31.3% (7.4%—55.9%)
Saint Lucia	455(442—468)	1,143(932—1,435)	151.2% (104.4%—213.6%)	8,703(8,459—8,957)	8,411(6,859—10,563)	−3.3% (−21.4%—20.7%)
Saint Vincent and the Grenadines	356(340—371)	631(578—690)	77.0% (60.3%—96.1%)	8,169(7,798—8,498)	7,927(7,264—8,678)	−3.0% (−12.1%—7.5%)
Suriname	882(854—911)	2,449(1,951—3,369)	177.8% (119.5%—283.9%)	7,219(6,991—7,457)	7,361(5,863—10,125)	2.0% (−19.4%—40.9%)
Trinidad and Tobago	3,758(3,679—3,831)	8,838(6,786—11,462)	135.2% (81.6%—204.4%)	7,960(7,793—8,116)	7,899(6,065—10,243)	−0.8% (−23.4%—28.4%)
Bermuda	276(268—285)	531(474—617)	92.4% (71.2%—124.3%)	7,948(7,708—8,201)	5,896(5,264—6,853)	−25.8% (−34%—−13.5%)
Puerto Rico	13,904(13,692—14,133)	24,392(21,053—27,718)	75.4% (51.1%—99.0%)	6,221(6,126—6,323)	4,631(3,997—5,262)	−25.6% (−35.9%—−15.5%)
Saint Kitts and Nevis	256(247—265)	241(217—262)	−5.7% (−16.3%—3.5%)	9,476(9,146—9,817)	8,688(7,792—9,434)	−8.3% (−18.6%—0.6%)
United States Virgin Islands	255(230—278)	608(483—794)	138.1% (83.8%—215.1%)	6,372(5,745—6,942)	4,713(3,747—6,156)	−26.0% (−42.9%—−2.1%)
Central Latin America	261,127(260,264—262,044)	1,009,392(963,365—1,059,754)	286.6% (269.3%—306.1%)	6,464(6,443—6,487)	7,389(7,052—7,757)	14.3% (9.2%—20.1%)
Colombia	50,879(50,585—51,168)	204,595(182,061—227,045)	302.1% (258.1%—346.1%)	6,082(6,047—6,117)	6,420(5,713—7,125)	5.6% (−6.0%—17.1%)
Costa Rica	5,228(5,084—5,382)	16,631(16,253—17,017)	218.1% (206.5%—229.2%)	5,519(5,367—5,682)	5,303(5,183—5,426)	−3.9% (−7.4%—−0.6%)
El Salvador	8,909(8,803—9,015)	27,707(24,284—31,207)	211.0% (172.8%—251.8%)	5,623(5,556—5,690)	7,236(6,342—8,150)	28.7% (12.9%—45.6%)
Guatemala	12,890(12,788—12,997)	49,243(45,010—53,771)	282.0% (249.6%—315.7%)	8,785(8,716—8,858)	8,354(7,636—9,122)	−4.9% (−13.0%—3.5%)
Honduras	5,600(5,146—6,078)	32,772(30,188—36,325)	485.2% (416.8%—564.2%)	5,740(5,274—6,229)	10,399(9,579—11,527)	81.2% (60.0%—105.6%)
Mexico	140,947(140,470—141,443)	512,158(510,792—513,593)	263.4% (261.8%—264.9%)	6,787(6,764—6,811)	7,580(7,560—7,601)	11.7% (11.2%—12.2%)
Nicaragua	3,285(3,214—3,357)	17,514(16,099—19,251)	433.1% (389.1%—486.3%)	4,464(4,368—4,561)	7,015(6,448—7,710)	57.1% (44.2%—72.8%)
Panama	4,050(3,961—4,150)	13,390(11,193—15,472)	230.6% (176.1%—284.6%)	5,018(4,909—5,143)	5,131(4,289—5,929)	2.2% (−14.6%—18.9%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Venezuela (Bolivarian Republic of)	29,338(29,036—29,646)	135,380(116,498—156,457)	361.4% (297.9%—432.8%)	6,180(6,116—6,245)	8,430(7,254—9,742)	36.4% (17.6%—57.5%)
Tropical Latin America	282,377(281,148—283,698)	901,596(895,970—907,847)	219.3% (216.8%—221.8%)	6,467(6,439—6,497)	6,277(6,238—6,321)	−2.9% (−3.7%—−2.2%)
Brazil	276,041(274,938—277,243)	876,197(871,065—881,691)	217.4% (215.0%—219.9%)	6,495(6,469—6,523)	6,231(6,195—6,270)	−4.1% (−4.8%—−3.3%)
Paraguay	6,336(5,813—6,912)	25,399(22,514—28,923)	300.9% (241.2%—367.1%)	5,464(5,013—5,961)	8,413(7,458—9,581)	54.0% (31.1%—79.4%)
Algeria	39,263(35,581—43,274)	142,995(127,330—158,702)	264.2% (212.2%—318.2%)	6,304(5,713—6,948)	8,265(7,359—9,173)	31.1% (12.4%—50.5%)
Bahrain	645(621—671)	2,210(1,991—2,430)	242.5% (208.9%—279.3%)	11,279(10,853—11,727)	8,772(7,905—9,645)	−22.2% (−29.8%—−13.9%)
Egypt	125,609(125,206—126,020)	281,344(249,323—319,234)	124.0% (98.6%—154.0%)	12,779(12,738—12,821)	12,393(10,982—14,062)	−3.0% (−14.0%—10.0%)
Iran (Islamic Republic of)	63,199(61,580—64,748)	303,829(297,568—310,040)	380.8% (365.9%—398.8%)	6,513(6,347—6,673)	8,320(8,149—8,490)	27.7% (23.8%—32.5%)
Iraq	30,839(27,360—34,410)	93,925(80,446—105,820)	204.6% (154.8%—258.8%)	7,720(6,849—8,614)	9,470(8,111—10,669)	22.7% (2.6%—44.5%)
Jordan	3,591(3,259—4,009)	19,301(16,683—22,012)	437.5% (347.1%—535.5%)	7,322(6,645—8,173)	6,285(5,432—7,168)	−14.2% (−28.6%—1.5%)
Kuwait	1,073(1,027—1,117)	5,365(4,694—6,083)	400.2% (334.9%—474.5%)	5,371(5,143—5,591)	5,434(4,755—6,161)	1.2% (−12.1%—16.2%)
Lebanon	7,933(7,269—9,039)	34,606(31,997—37,939)	336.2% (275.6%—397.6%)	8,295(7,601—9,452)	8,999(8,321—9,866)	8.5% (−6.6%—23.8%)
Libya	5,636(4,931—6,385)	19,284(16,536—22,407)	242.2% (184.6%—314%)	6,485(5,674—7,347)	8,800(7,546—10,225)	35.7% (12.8%—64.2%)
Morocco	55,180(52,070—58,818)	148,076(127,074—160,503)	168.4% (128.9%—197.4%)	7,779(7,340—8,292)	8,960(7,690—9,712)	15.2% (−1.8%—27.7%)
Palestine	4,332(3,730—4,978)	8,661(7,844—9,472)	99.9% (70.2%—134.8%)	10,272(8,843—11,804)	8,343(7,556—9,124)	−18.8% (−30.8%—−4.6%)
Oman	2,319(1,890—2,748)	5,833(5,229—6,492)	151.5% (106.5%—215.2%)	8,035(6,548—9,522)	8,447(7,574—9,402)	5.1% (−13.7%—31.8%)
Qatar	295(263—326)	1,171(1,000—1,380)	297.2% (229.4%—384.1%)	12,217(10,897—13,510)	6,669(5,692—7,859)	−45.4% (−54.7%—−33.5%)
Saudi Arabia	21,463(18,076—24,777)	34,776(31,432—38,843)	62.0% (38.2%—96.3%)	8,646(7,282—9,981)	6,701(6,057—7,485)	−22.5% (−33.9%—−6.1%)
Syrian Arab Republic	20,028(17,428—22,860)	50,153(41,732—59,685)	150.4% (100.2%—215.8%)	8,990(7,823—10,261)	8,493(7,067—10,107)	−5.5% (−24.5%—19.1%)
Tunisia	14,975(13,719—16,336)	59,512(49,470—70,239)	297.4% (223.1%—386.9%)	6,194(5,675—6,757)	8,547(7,105—10,088)	38.0% (12.2%—69.1%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Turkey	129,271(121,388—136,059)	396,190(348,410—446,061)	206.5% (169.1%—246.9%)	8,618(8,092—9,070)	7,793(6,853—8,774)	−9.6% (−20.6%—2.3%)
United Arab Emirates	1,059(900—1,226)	4,042(3,512—4,579)	281.6% (243.9%—321.7%)	8,704(7,395—10,071)	8,372(7,275—9,486)	−3.8% (−13.3%—6.3%)
Yemen	17,204(14,719—20,203)	61,708(49,620—76,562)	258.7% (184.1%—358.0%)	8,819(7,545—10,356)	10,374(8,342—12,871)	17.6% (−6.8%—50.2%)
Afghanistan	35,997(30,539—40,594)	60,300(52,749—67,410)	67.5% (44.1%—99.2%)	11,583(9,827—13,062)	14,906(13,039—16,663)	28.7% (10.7%—53.0%)
Sudan	39,583(35,154—44,824)	83,813(69,920—100,642)	111.7% (72.9%—162.0%)	8,399(7,459—9,511)	9,789(8,167—11,755)	16.6% (−4.8%—44.2%)
North Africa and Middle East	619,832(598,128—641,674)	1,818,792(1,695,620—1,941,980)	193.4% (175.8%—211.3%)	8,583(8,282—8,885)	8,945(8,339—9,550)	4.2% (−2.1%—10.6%)
Bangladesh	203,851(190,431—217,726)	540,577(465,412—621,027)	165.2% (125.2%—208.5%)	8,959(8,369—9,569)	7,335(6,315—8,427)	−18.1% (−30.5%—−4.8%)
Bhutan	755(663—865)	2,306(2,069—2,478)	205.5% (162%—255.9%)	8,418(7,394—9,650)	7,153(6,419—7,687)	−15.0% (−27.1%—−1.0%)
India	1,589,680(1,536,532—1,644,431)	5,248,722(4,981,205—5,530,121)	230.2% (209.5%—252.2%)	8,826(8,531—9,130)	8,809(8,360—9,281)	−0.2% (−6.4%—6.5%)
Nepal	34,590(30,523—39,334)	114,748(103,836—130,483)	231.7% (182.2%—298.7%)	9,487(8,371—10,788)	9,849(8,913—11,200)	3.8% (−11.7%—24.8%)
Pakistan	230,581(220,447—240,363)	510,949(458,977—581,061)	121.6% (95.7%—155.8%)	8,156(7,798—8,502)	10,090(9,064—11,475)	23.7% (9.3%—42.8%)
South Asia	2,059,456(2,003,387—2,119,402)	6,417,303(6,128,363—6,712,493)	211.6% (193.9%—229.4%)	8,768(8,530—9,023)	8,765(8,370—9,168)	0.0% (−5.7%—5.7%)
Southeast Asia, East Asia, and Oceania	4,303,427(4,010,954—4,592,670)	10,351,445(9,313,442—11,361,424)	140.5% (110.5%—174.5%)	8,618(8,032—9,197)	6,725(6,050—7,381)	−22.0% (−31.7%—−10.9%)
East Asia	3,413,063(3,128,410—3,703,625)	7,918,054(6,896,007—8,947,951)	132% (98.2%—174.1%)	8,771(8,039—9,518)	6,407(5,580—7,241)	−26.9% (−37.6%—−13.7%)
China	3,313,720(3,027,994—3,605,380)	7,672,921(6,653,672—8,692,054)	131.6% (96.7%—174.8%)	8,830(8,068—9,607)	6,431(5,577—7,286)	−27.2% (−38.1%—−13.6%)
Democratic People's Republic of Korea	54,012(45,383—61,190)	125,273(110,897—141,498)	131.9% (95.7%—179.1%)	8,249(6,931—9,345)	7,158(6,337—8,085)	−13.2% (−26.8%—4.4%)
Taiwan (Province of China)	45,331(44,997—45,683)	119,861(118,946—121,062)	164.4% (161.2%—167.4%)	6,216(6,170—6,264)	4,746(4,710—4,794)	−23.6% (−24.6%—−22.8%)
Oceania	9,260(8,163—10,480)	28,258(22,996—35,114)	205.2% (142.3%—289.6%)	8,907(7,852—10,081)	10,244(8,336—12,729)	15.0% (−8.7%—46.8%)
Viet Nam	164,467(149,653—192,225)	347,702(302,716—383,006)	111.4% (77.3%—142.1%)	8,066(7,339—9,427)	7,783(6,776—8,573)	−3.5% (−19.1%—10.5%)
Fiji	1,226(1,124—1,353)	3,507(2,742—4,418)	186.2% (121.4%—275.8%)	9,063(8,311—10,004)	10,901(8,524—13,732)	20.3% (−6.9%—57.9%)
Kiribati	153(136—165)	243(220—273)	59.2% (40.7%—84.8%)	10,601(9,421—11,431)	9,779(8,861—10,967)	−7.8% (−18.5%—7.1%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Marshall Islands	76(72—80)	102(87—119)	33.3% (12.8%—56.1%)	10,390(9,814—10,931)	8,854(7,552—10,406)	−14.8% (−27.9%—−0.2%)
Micronesia (Federated States of)	245(225—272)	230(199—271)	−6.2% (−20.6%—12.4%)	10,447(9,573—11,564)	9,111(7,861—10,744)	−12.8% (−26.1%—4.5%)
Papua New Guinea	5,299(4,400—6,291)	18,986(14,485—24,734)	258.3% (161.9%—386.2%)	8,826(7,329—10,477)	10,910(8,323—14,213)	23.6% (−9.6%—67.7%)
Samoa	346(320—369)	567(515—640)	63.8% (50.0%—85.6%)	9,135(8,437—9,741)	8,297(7,537—9,364)	−9.2% (−16.8%—2.9%)
Solomon Islands	486(441—552)	1,253(1,095—1,480)	158.0% (123.0%—202.4%)	10,348(9,389—11,752)	9,456(8,265—11,168)	−8.6% (−21.0%—7.1%)
Tonga	185(169—202)	338(296—385)	82.9% (54.1%—116.2%)	7,119(6,499—7,783)	7,721(6,750—8,793)	8.4% (−8.6%—28.2%)
Vanuatu	219(200—250)	661(575—782)	201.2% (156.2%—261.1%)	9,533(8,672—10,874)	9,926(8,637—11,742)	4.1% (−11.4%—24.9%)
American Samoa	67(65—69)	174(159—195)	160.6% (136.3%—192.6%)	7,599(7,350—7,834)	7,850(7,195—8,798)	3.3% (−6.3%—16.0%)
Cook Islands	52(50—54)	98(87—110)	89.6% (65.5%—114.2%)	8,814(8,432—9,194)	6,314(5,595—7,090)	−28.4% (−37.5%—−19.1%)
Guam	172(166—178)	520(456—571)	202.9% (164.8%—237.5%)	6,167(5,946—6,389)	4,250(3,730—4,669)	−31.1% (−39.7%—−23.2%)
Nauru	17(15—19)	24(20—27)	39.2% (18.2%—63.8%)	10,742(9,502—11,966)	10,430(9,007—12,018)	−2.9% (−17.6%—14.3%)
Niue	16(14—17)	11(10—12)	−28.5% (−37%—−19.7%)	10,609(9,638—11,542)	9,381(8,390—10,022)	−11.6% (−22.1%—−0.7%)
Northern Mariana Islands	33(30—38)	151(125—171)	356.8% (259.5%—445.4%)	6,879(6,208—7,984)	7,130(5,908—8,088)	3.7% (−18.4%—23.7%)
Palau	43(38—48)	73(62—85)	70.0% (42.6%—102.7%)	9,003(7,943—10,125)	8,021(6,828—9,311)	−10.9% (−25.3%—6.2%)
San Marino	138(126—151)	207(166—255)	50.3% (19.7%—89.9%)	5,906(5,390—6,480)	3,979(3,195—4,889)	−32.6% (−46.3%—−14.9%)
Tokelau	7(7—8)	7(6—8)	2.3% (−11.1%—17.5%)	9,579(8,754—10,497)	8,059(6,843—9,225)	−15.9% (−26.9%—−3.4%)
Tuvalu	28(25—30)	43(40—49)	57.4% (43.8%—75.1%)	10,233(9,414—10,930)	8,536(7,805—9,588)	−16.6% (−23.8%—−7.2%)
Southeast Asia	881,104(846,528—920,203)	2,405,132(2,146,290—2,673,326)	173% (144.6%—204.4%)	8,071(7,754—8,429)	7,996(7,136—8,888)	−0.9% (−11.2%—10.5%)
Cambodia	18,888(17,265—20,665)	50,441(41,638—58,338)	167.1% (116.4%—217.5%)	10,128(9,258—11,081)	9,220(7,611—10,664)	−9.0% (−26.2%—8.2%)
Indonesia	319,923(304,701—335,864)	872,354(708,623—1,047,487)	172.7% (121.9%—227.8%)	8,401(8,001—8,820)	9,044(7,346—10,859)	7.6% (−12.4%—29.4%)
Lao People's Democratic Republic	9,521(8,236—10,763)	17,664(14,576—20,745)	85.5% (48.2%—129.7%)	10,729(9,281—12,129)	9,170(7,567—10,769)	−14.5% (−31.7%—5.8%)
Malaysia	32,053(31,810—32,302)	109,456(103,258—117,450)	241.5% (222.4%—266.1%)	7,366(7,310—7,423)	7,740(7,302—8,306)	5.1% (−0.8%—12.7%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Maldives	243(233—253)	809(700—931)	233.4% (188.5%—285.5%)	8,092(7,761—8,443)	5,951(5,147—6,846)	−26.5% (−36.4%—−15.0%)
Myanmar	104,244(89,803—118,836)	213,954(177,016—257,076)	105.2% (60.4%—161.7%)	10,658(9,182—12,150)	9,574(7,921—11,504)	−10.2% (−29.8%—14.5%)
Philippines	86,311(82,175—89,988)	343,920(313,290—370,843)	298.5% (261.9%—335.6%)	6,660(6,341—6,944)	9,507(8,660—10,251)	42.8% (29.6%—56.1%)
Sri Lanka	39,199(37,622—40,966)	91,337(65,081—117,114)	133.0% (64.8%—199.9%)	7,949(7,629—8,307)	6,003(4,277—7,697)	−24.5% (−46.6%—−2.8%)
Thailand	101,314(94,223—109,360)	343,060(276,218—412,517)	238.6% (171.8%—311.8%)	6,628(6,164—7,154)	5,506(4,434—6,621)	−16.9% (−33.3%—1.0%)
Timor-Leste	770(692—868)	3,873(3,130—4,642)	403.3% (293.8%—534.0%)	9,020(8,105—10,168)	8,566(6,923—10,266)	−5.0% (−25.7%—19.6%)
Mauritius	2,621(2,559—2,680)	6,767(6,174—7,350)	158.2% (134%—182.5%)	7,959(7,770—8,139)	6,748(6,157—7,330)	−15.2% (−23.2%—−7.2%)
Seychelles	276(267—286)	442(415—472)	60% (48.5%—71.9%)	8,068(7,805—8,360)	7,996(7,521—8,542)	−0.9% (−8.0%—6.5%)
Sub-Saharan Africa	863,387(826,317—900,392)	2,183,660(2,051,825—2,309,196)	152.9% (138.4%—168.5%)	9,342(8,941—9,742)	11,146(10,473—11,786)	19.3% (12.4%—26.7%)
Central Sub-Saharan Africa	80,716(72,734—88,934)	226,644(196,582—254,734)	180.8% (139.7%—230.1%)	9,957(8,973—10,971)	12,021(10,427—13,511)	20.7% (3.1%—41.9%)
Angola	14,179(12,293—16,044)	47,475(40,152—54,972)	234.8% (175.3%—305.5%)	11,056(9,586—12,510)	11,831(10,006—13,699)	7.0% (−12.0%—29.6%)
Central African Republic	4,570(4,212—4,826)	9,513(8,036—11,185)	108.2% (75.7%—145.8%)	12,299(11,336—12,989)	14,910(12,595—17,530)	21.2% (2.3%—43.2%)
Congo	4,903(4,509—5,303)	11,485(10,073—12,698)	134.2% (105.1%—166.0%)	11,245(10,342—12,163)	11,943(10,474—13,204)	6.2% (−7.0%—20.6%)
Democratic Republic of the Congo	53,196(47,310—59,403)	151,127(127,773—172,483)	184.1% (133.2%—244.6%)	9,401(8,360—10,498)	11,952(10,105—13,641)	27.1% (4.4%—54.2%)
Equatorial Guinea	881(762—978)	2,213(1,766—2,720)	151.1% (98.7%—217.7%)	11,744(10,156—13,029)	11,557(9,223—14,208)	−1.6% (−22.1%—24.5%)
Gabon	2,987(2,644—3,277)	4,831(4,223—5,372)	61.8% (37.6%—90.7%)	10,582(9,368—11,610)	11,923(10,422—13,257)	12.7% (−4.1%—32.9%)
Eastern Sub-Saharan Africa	326,412(314,263—337,442)	803,184(750,748—856,578)	146.1% (131.5%—162.4%)	10,480(10,090—10,835)	11,657(10,896—12,432)	11.2% (4.6%—18.6%)
Burundi	13,961(12,267—15,294)	19,725(16,978—22,903)	41.3% (19.2%—70.5%)	12,339(10,841—13,517)	10,950(9,425—12,714)	−11.3% (−25.1%—7.1%)
Comoros	765(661—882)	2,194(1,924—2,536)	186.9% (136.3%—249.7%)	9,439(8,155—10,885)	9,669(8,481—11,175)	2.4% (−15.6%—24.9%)
Djibouti	359(305—424)	2,203(1,864—2,658)	513.6% (397%—670.9%)	8,447(7,169—9,986)	10,059(8,511—12,139)	19.1% (−3.6%—49.6%)
Eritrea	3,305(3,076—3,535)	10,753(9,712—11,837)	225.4% (192.4%—262.8%)	11,735(10,921—12,552)	11,713(10,580—12,894)	−0.2% (−10.3%—11.3%)
Ethiopia	95,047(90,427—98,254)	212,037(197,511—230,029)	123.1% (106.0%—146%)	12,298(11,700—12,713)	10,755(10,018—11,667)	−12.6% (−19.2%—−3.6%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Kenya	29,985(28,116—31,788)	102,332(92,799—112,506)	241.3% (204.3%—283.9%)	8,092(7,588—8,579)	10,967(9,945—12,057)	35.5% (20.8%—52.5%)
Madagascar	22,749(21,155—24,255)	41,786(35,366—48,249)	83.7% (52.1%—115.6%)	10,597(9,855—11,299)	11,488(9,723—13,265)	8.4% (−10.3%—27.2%)
Malawi	15,480(14,526—16,436)	43,184(38,972—48,098)	179.0% (151.1%—208.2%)	9,953(9,339—10,568)	13,882(12,528—15,461)	39.5% (25.5%—54.1%)
Mozambique	24,073(22,450—26,213)	59,540(49,906—67,679)	147.3% (104.4%—184.7%)	9,905(9,237—10,786)	13,850(11,609—15,744)	39.8% (15.6%—61.0%)
Rwanda	13,560(12,616—14,371)	26,092(22,813—29,792)	92.4% (64.2%—126.4%)	12,280(11,425—13,015)	10,560(9,233—12,057)	−14.0% (−26.6%—1.2%)
Somalia	9,124(8,030—9,968)	30,375(25,558—35,043)	232.9% (180.3%—292.5%)	13,001(11,442—14,204)	15,943(13,415—18,393)	22.6% (3.2%—44.6%)
United Republic of Tanzania	43,552(40,158—47,348)	122,708(110,809—139,389)	181.7% (147.2%—228.2%)	8,940(8,243—9,719)	11,048(9,977—12,550)	23.6% (8.4%—43.9%)
Uganda	28,856(26,034—32,060)	69,850(58,389—80,455)	142.1% (101.3%—190.6%)	9,859(8,895—10,954)	11,608(9,704—13,371)	17.7% (−2.1%—41.4%)
Zambia	12,600(11,573—13,625)	41,418(34,613—47,590)	228.7% (171.9%—288.7%)	10,877(9,991—11,762)	15,392(12,863—17,686)	41.5% (17.1%—67.3%)
South Sudan	12,764(11,025—14,829)	18,288(15,574—21,373)	43.3% (17.3%—73.3%)	10,157(8,773—11,800)	13,138(11,188—15,355)	29.4% (5.9%—56.4%)
Southern Sub-Saharan Africa	86,037(82,905—89,408)	325,604(316,110—335,138)	278.4% (262.7%—295.3%)	6,628(6,386—6,887)	12,209(11,853—12,567)	84.2% (76.6%—92.4%)
Botswana	2,077(1,700—2,367)	8,042(6,917—9,014)	287.2% (216.6%—385.8%)	9,041(7,402—10,306)	13,735(11,814—15,394)	51.9% (24.2%—90.6%)
Lesotho	3,876(3,449—4,461)	9,502(8,247—10,630)	145.2% (102.0%—189.2%)	9,035(8,040—10,400)	19,764(17,154—22,110)	118.7% (80.2%—158%)
Namibia	2,419(2,200—2,632)	8,564(7,556—9,515)	254.1% (204.1%—308.5%)	9,109(8,285—9,911)	14,306(12,621—15,894)	57.1% (34.9%—81.2%)
South Africa	61,759(59,929—63,583)	249,595(244,218—254,656)	304.1% (289.8%—318.2%)	6,087(5,907—6,267)	11,281(11,038—11,510)	85.3% (78.7%—91.8%)
Zimbabwe	14,736(12,791—16,311)	45,773(40,379—50,996)	210.6% (165.4%—271.9%)	8,217(7,133—9,095)	17,243(15,211—19,210)	109.8% (79.3%—151.3%)
Western Sub-Saharan Africa	370,222(347,318—393,279)	828,228(773,635—893,006)	123.7% (103.7%—143.4%)	9,212(8,642—9,786)	10,163(9,493—10,958)	10.3% (0.5%—20.0%)
Eswatini	1,170(1,027—1,311)	4,128(3,366—4,713)	252.6% (177.6%—328.7%)	9,900(8,686—11,090)	18,359(14,969—20,963)	85.4% (46.0%—125.5%)
Benin	10,155(9,504—10,883)	23,207(21,216—25,719)	128.5% (104.7%—157.4%)	9,899(9,265—10,609)	10,369(9,480—11,492)	4.7% (−6.2%—18.0%)
Burkina Faso	18,667(17,193—20,380)	46,632(41,597—52,764)	149.8% (117.4%—188.3%)	9,508(8,758—10,381)	11,071(9,876—12,527)	16.4% (1.3%—34.4%)

Table 2 (continued)

	1990 death counts (95% UI)	2021 death counts (95% UI)	Annual percentage change of death counts (95% CI)	1990 death rates (95% UI)	2021 death rates (95% UI)	Annual percentage change of death rates (95% CI)
Cameroon	17,081(15,739—18,513)	58,305(51,535—69,919)	241.3% (195.5%—307.5%)	9,278(8,549—10,056)	11,654(10,301—13,976)	25.6% (8.8%—50.0%)
Cabo Verde	974(884—1,066)	1,889(1,665—2,066)	94.0% (68.2%—122.3%)	6,511(5,914—7,129)	9,222(8,130—10,086)	41.6% (22.7%—62.3%)
Chad	14,521(13,574—15,669)	26,650(23,104—31,154)	83.5% (56.8%—114.3%)	9,641(9,012—10,404)	11,182(9,694—13,072)	16.0% (−0.9%—35.4%)
Côte d'Ivoire	12,175(11,118—13,372)	41,594(37,171—48,708)	241.6% (196.5%—302.5%)	9,099(8,310—9,994)	9,975(8,915—11,682)	9.6% (−4.9%—29.2%)
Gambia	1,350(1,177—1,551)	5,296(4,413—5,969)	292.3% (217.0%—375.8%)	8,751(7,630—10,051)	11,814(9,844—13,316)	35.0% (9.1%—63.7%)
Ghana	22,800(20,238—25,929)	71,367(61,438—81,301)	213.0% (158.3%—273.8%)	8,998(7,987—10,232)	10,367(8,925—11,810)	15.2% (−4.9%—37.6%)
Guinea	16,056(14,818—17,384)	31,587(27,078—36,641)	96.7% (67.9%—129.2%)	9,683(8,936—10,484)	12,171(10,434—14,119)	25.7% (7.3%—46.4%)
Guinea-Bissau	1,962(1,735—2,178)	3,371(2,822—3,856)	71.9% (41.1%—100.6%)	11,994(10,608—13,315)	13,280(11,115—15,189)	10.7% (−9.1%—29.2%)
Liberia	5,310(4,784—5,932)	8,565(7,434—10,114)	61.3% (35.2%—95.4%)	8,925(8,040—9,971)	10,740(9,321—12,683)	20.3% (0.8%—45.8%)
Mali	15,760(14,611—17,014)	42,944(38,470—48,044)	172.5% (139.6%—211.7%)	9,699(8,992—10,471)	11,632(10,420—13,014)	19.9% (5.4%—37.2%)
Mauritania	4,953(4,411—5,507)	9,311(7,885—11,084)	88.0% (55.3%—128.8%)	9,493(8,454—10,555)	9,146(7,746—10,889)	−3.6% (−20.4%—17.3%)
Niger	9,854(9,163—10,585)	32,995(28,773—38,176)	234.9% (189.8%—297.5%)	9,585(8,913—10,296)	9,684(8,445—11,204)	1.0% (−12.6%—19.9%)
Nigeria	188,953(169,643—209,375)	353,698(327,285—390,105)	87.2% (64.4%—114.1%)	9,019(8,097—9,993)	9,446(8,741—10,419)	4.7% (−8.0%—19.8%)
Sao Tome and Principe	284(265—304)	434(416—452)	52.7% (40.8%—64.3%)	8,602(8,026—9,203)	9,465(9,080—9,869)	10.0% (1.5%—18.4%)
Senegal	14,020(13,059—15,281)	38,617(33,434—44,893)	175.4% (132.8%—222.1%)	9,271(8,636—10,105)	10,907(9,443—12,679)	17.6% (−0.6%—37.6%)
Sierra Leone	10,654(9,553—11,773)	17,217(14,965—19,869)	61.6% (37.9%—93.1%)	9,862(8,843—10,898)	10,041(8,727—11,588)	1.8% (−13.1%—21.7%)
Togo	4,682(4,280—5,175)	14,539(12,347—17,277)	210.6% (162.2%—272.9%)	9,234(8,442—10,207)	10,059(8,543—11,954)	8.9% (−8%—30.8%)

Abbreviations: CI confidence interval, *SDI* socio-demographic index, *UI* uncertainty intervals

and occupational risks tend to decrease as SDI increases, while metabolic risks show a positive correlation with SDI.

Discussion

In 2021, the global population aged 70 and older was approximately 494.4 million. Despite declines in mortality and disability rates, the global death and DALYs absolute numbers more than doubled from 1990 to 2021. Regions with low and low-middle SDI levels experienced disproportionately higher disease burdens than high SDI regions. Ischemic heart disease, stroke, and COVID-19 emerged as the leading causes of death among older adults globally in 2021. Meanwhile, the primary risk factors contributing to DALYs included high systolic blood pressure, air pollution, and tobacco use. These findings highlighted the persistent challenges and health disparities faced by aging populations worldwide, influenced by

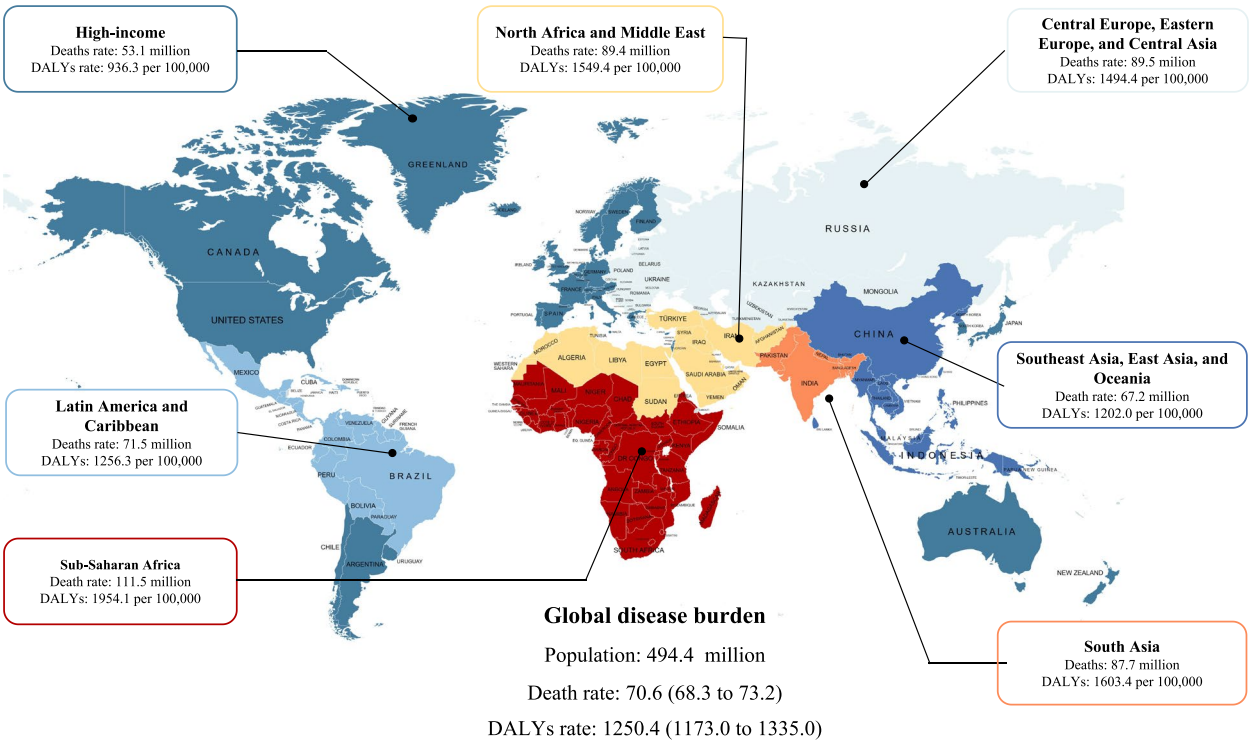

Fig. 1 Global and regional burdens of disease in adults aged ≥ 70 in 2021. *Abbreviations:* DALYs, deaths and disability-adjusted life years

both epidemiological transitions and varying socio-economic contexts.

Rapidly Increased Global Burden in Adults aged 70 and older

The growing global population of adults aged 70 is increasingly vulnerable to a higher prevalence of chronic diseases and disabilities typically associated with aging. While global mortality rates generally declined from 1990 to 2019, the emergence of COVID-19 has notably reversed this trend, resulting in a noticeable increase in disease burden among this age group from 2019 to 2021. Our analysis identifies COVID-19, along with conditions like ischemic cardiomyopathy and stroke, as the third leading cause of DALYs in this demographic. Looking forward, ongoing advancements in medical treatments, disease prevention strategies, and improvements in public health infrastructure are expected to sustain a downward trajectory in disease burden over the long term. As people age and life expectancy continues to rise in many parts of the world, the older population becomes increasingly susceptible to developing chronic conditions such as cardiovascular diseases, diabetes, arthritis, and cognitive decline. There are significant disparities in disease burden among regions with low and low-middle SDI levels compared to those with high SDI. These inequities arise from several factors, including variations in population aging and disparities in healthcare infrastructure, which limit access to comprehensive healthcare services encompassing both preventive and therapeutic interventions for chronic diseases [21]. Regions with lower SDI experience higher prevalence rates of risk factors such as substandard sanitation, increased exposure to environmental pollutants, and inadequate nutrition [22–24]. Socioeconomic factors such as poverty, lack of education, and inequality further exacerbate these health disparities. The aging population presents unique global public health challenges, necessitating proactive healthcare policies that address their complex needs. Strategies such as emphasizing prevention, early detection, and effective chronic disease management are crucial to mitigate their impact. Addressing social determinants of health, such as income inequality, education, housing conditions, and environmental factors, is essential to enhance overall well-being and reduce disparities.

Regional Variations of Risk Factors Among Adults aged 70 and older

High blood pressure is a major risk factor for cardiovascular diseases, stroke, Alzheimer's Disease and other health complications [25–28]. It is a global health concern and contributes significantly to the global burden of disease among the older population. Effective blood pressure management and control through lifestyle

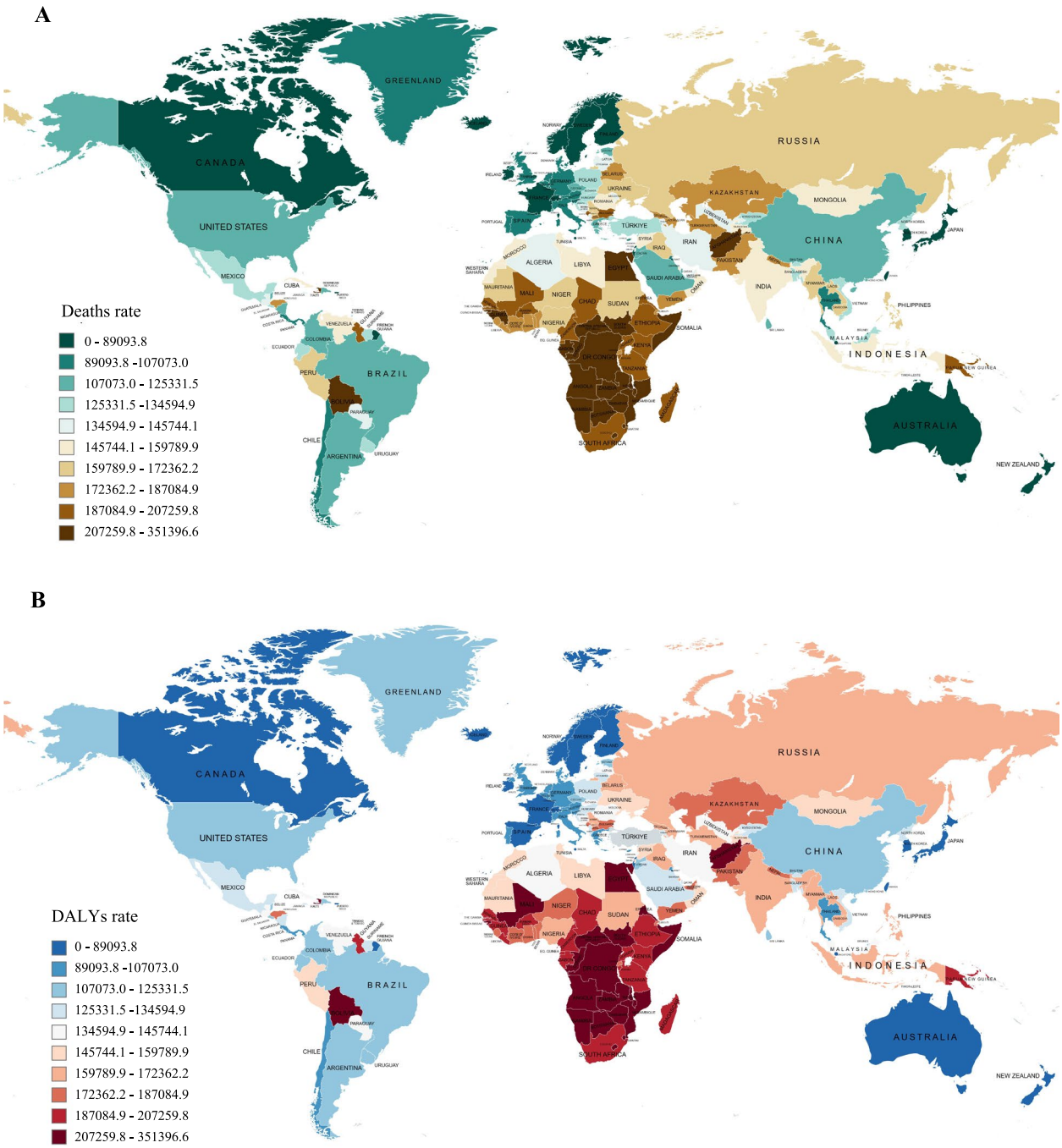

Fig. 2 Death and DALY rates of adults aged ≥ 70 among 204 countries and territories in 2021: (**A**) Death rates; **B** DALY rates. *Abbreviations:* DALYs, deaths and disability-adjusted life years

modifications and medication are crucial in reducing associated health risks [29]. Drawing on the GBD data, we found that air pollution exhibits regional disparities in its distribution and impact, particularly affecting South Asia, Sub-Saharan Africa, and Southeast Asia, East Asia, and Oceania. These regions experience high levels of air pollution due to rapid industrialization, urbanization, and population growth [30]. Governments and international organizations are increasingly focusing on mitigating air pollution through policies to reduce emissions, promote cleaner technologies, and improve urban planning. The disparities between males and females in

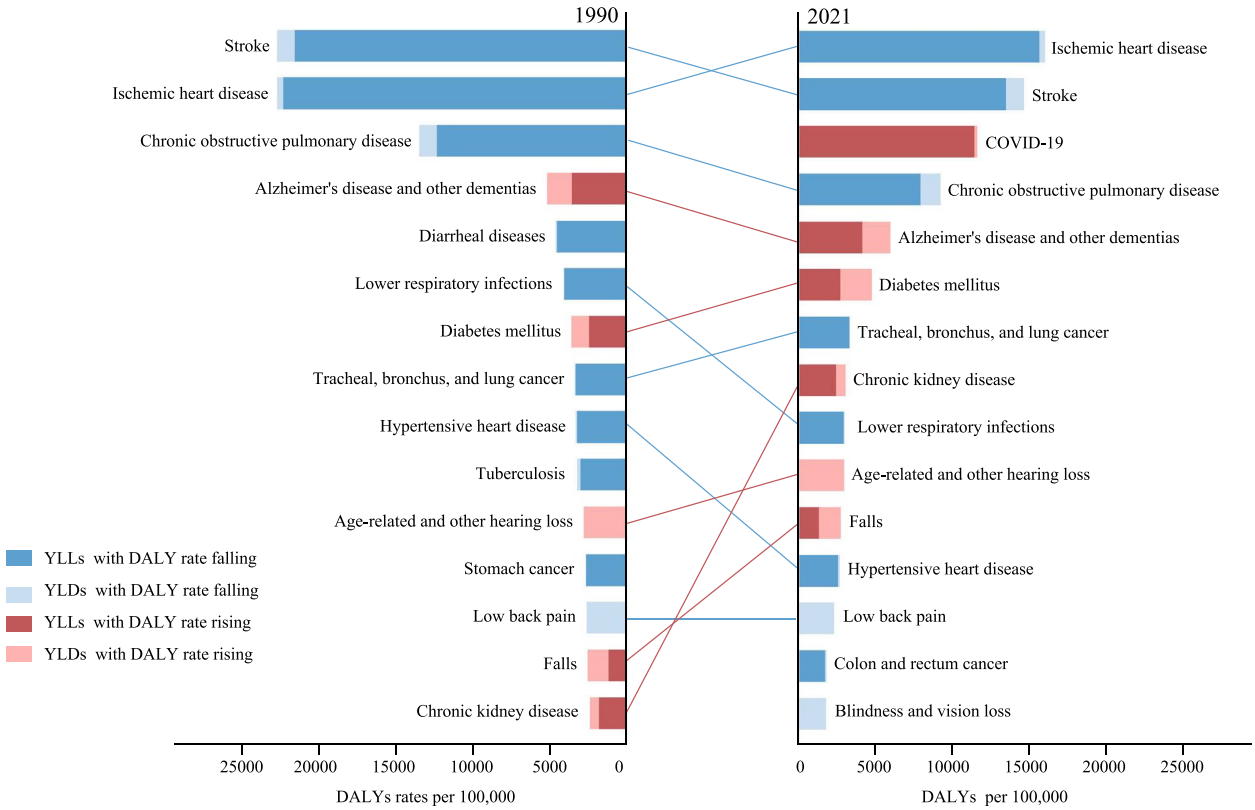

Fig. 3 Temporal changes in the top 15 causes of total DALYs rates globally among adults aged ≥ 70 between 1990 and 2021. This figure illustrates the evolution of the top 15 causes contributing to DALY rates among adults aged 70 and older globally, comparing data from 1990 and 2021. Each health condition is ranked based on its DALY rates per 100,000 population in these respective years. The color scheme indicates whether the DALY rates were higher in 2021 (red) or 1990 (blue), determined by whether the 95% uncertainty interval of the absolute difference in DALY rates includes zero. The transparency of the bars and lines reflects the distribution of DALYs between Years of Life Lost (YLL) and Years Lived with Disability (YLD) for each cause. *Abbreviations:* DALY, disability-adjusted life-year; YLD, years lived with disability; YLL, years of life lost

terms of health risks among older populations are striking and necessitate tailored public health strategies. Tobacco remains one of the top three contributors to disease burden among older men [31]. This habit leads to a range of health issues, including cardiovascular diseases, respiratory disorders, and cancers [31, 32]. The behavior of smoking is highly modifiable, presenting a clear opportunity for policymakers to prioritize targeted smoking cessation programs [33]. These programs can effectively reduce smoking rates among older men, thereby lowering associated health risks and improving overall health outcomes [34].

Global Burden Trends in Older Populations: Insights on Metabolic Risk Factors

Metabolic risk factors such as high body mass index and high fasting plasma glucose significantly contribute to the DALYs of older population, especially among older women [35, 36]. Addressing metabolic risks requires targeted interventions that promote healthier lifestyles, increase awareness about metabolic health, and improve access to healthcare services [37, 38]. These efforts are crucial in mitigating the impact of metabolic disorders and enhancing the quality of life for older women [39, 40]. The correlation between SDI and SEV of risk factors among older populations reveals significant insights into global health trends. Regions with lower SDI scores tend to face a higher SEV of environmental and behavioral risk factors among older adults. These factors can include issues like exposure to pollution, limited access to healthcare, and higher rates of smoking and poor diet choices [41–43]. Conversely, regions with higher SDI scores tend to show a greater impact on the SEV of metabolic factors such as high fasting plasma glucose and high body mass index [36, 44]. These conditions are often associated with lifestyle changes, including sedentary behavior and diets high in processed foods, which are more prevalent in higher-income settings. The analysis also highlights a concerning trend in DALYs among older adults. While overall DALYs rates for many risk factors have decreased from 1990 to 2021, there has been a noticeable increase in DALYs attributed to metabolic factors. While overall

	Male					Female				
	1	2	3	4	5	1	2	3	4	5
Global	IHD 18669.8	Stroke 16532.5	COVID-19 15069.2	COPD 11399.6	DM 4992.9	IHD 13895.2	Stroke 13132.3	COVID-19 8846.6	COPD 7454.5	AD 7165
Central Europe, Eastern Europe, and Central Asia	IHD 39543.5	COVID-19 30779.4	Stroke 20412.7	Tracheal, bronchus, and lung cancer 5810.3	COPD 4784	IHD 31716.1	COVID-19 19526.9	Stroke 18396	AD 6121.9	Low back pain 4025
Central Asia	IHD 52100	COVID-19 28267.3	Stroke 23887.7	COPD 5997.9	AD 3812.7	IHD 39494.4	Stroke 20210.3	COVID-19 17341.5	AD 5658	DM 3973.6
Central Europe	IHD 26823	COVID-19 26075.5	Stroke 17823.1	Tracheal, bronchus, and lung cancer 7187.6	COPD 5005.5	IHD 20574.8	Stroke 15847.3	COVID-19 14943	AD 6114.9	HHD 4513.5
Eastern Europe	IHD 47988.8	COVID-19 35278.2	Stroke 21955.2	Tracheal, bronchus, and lung cancer 5159.7	COPD 4361.5	IHD 37531.1	COVID-19 22695	Stroke 19721.2	AD 6193	Low back pain 4204.8
High-income	IHD 10667.4	COVID-19 7776.2	Stroke 6397.5	Tracheal, bronchus, and lung cancer 5405.4	COPD 5378.2	AD 8975.1	IHD 6808.4	Stroke 5972.8	COVID-19 4980.5	COPD 3832
Australasia	IHD 9769.9	Stroke 4806.3	COPD 4520.1	AD 4220.9	Tracheal, bronchus, and lung cancer 4136.7	AD 7603.9	IHD 6331.4	Stroke 4975.2	Falls 4463.2	COPD 3584.3
High-income Asia Pacific	Stroke 7753.3	Tracheal, bronchus, and lung cancer 6158.2	IHD 5485.2	AD 5235.8	Lower respiratory infections 4295.3	AD 9957.5	Stroke 5942	IHD 3723	Low back pain 2949.4	DM 2664.6
High-income North America	IHD 14646.8	COVID-19 11500.3	COPD 7291.1	DM 5742.7	Stroke 5607	IHD 9373.7	AD 8516.2	COVID-19 8108.5	COPD 6561.9	Stroke 6028.7
Southern Latin America	COVID-19 21906.4	IHD 10676.2	Stroke 8267.1	Lower respiratory infections 7140.5	COPD 5642.2	COVID-19 13112.2	IHD 6804.1	Stroke 6567.8	AD 6354.7	Lower respiratory infections 5703.7
Western Europe	IHD 10757.8	COVID-19 8044.9	Stroke 6166.3	AD 5448.8	Tracheal, bronchus, and lung cancer 5309	AD 9044	IHD 6827.8	Stroke 5953.5	COVID-19 4864.3	Falls 4131.3
Latin America and Caribbean	COVID-19 31888.2	IHD 15077.7	Stroke 8202.7	DM 7600.3	COPD 5855.1	COVID-19 17761.1	IHD 10637	DM 7733.9	Stroke 6925.5	AD 6054.1
Andean Latin America	COVID-19 56290.8	IHD 10694.9	Lower respiratory infections 8198.8	Stroke 6242.9	CKD 6018.3	COVID-19 31250.2	IHD 8232.1	Lower respiratory infections 7173.1	DM 6174.9	Stroke 6110.6
Caribbean	COVID-19 28419.5	IHD 18892.8	Stroke 11396.3	DM 7927.6	Prostate cancer 7530.8	COVID-19 15644.2	IHD 15023.6	Stroke 10657.7	DM 8836.4	AD 4963.5
Central Latin America	COVID-19 32402.3	IHD 19076.5	DM 9198	CKD 6649.8	Stroke 6510.6	COVID-19 18432.3	IHD 13711.1	DM 9443.5	Stroke 5883.6	CKD 5250.4
Tropical Latin America	COVID-19 26062.2	IHD 11251.1	Stroke 9639.5	DM 6403.6	COPD 6235.5	COVID-19 14743.4	AD 7524.4	IHD 7414.6	Stroke 7255.5	DM 6272.4
North Africa and Middle East	IHD 31351.4	COVID-19 29725.8	Stroke 16450.7	DM 7501.4	COPD 6256.5	IHD 27747.5	Stroke 17147.5	COVID-19 15686.8	DM 8106.9	HHD 6755.3
South Asia	COVID-19 27125	IHD 24177	COPD 21722.7	Stroke 13523.3	DM 6675.5	COPD 17315.6	IHD 16124.9	COVID-19 14655.8	Stroke 11423.8	Diarrheal diseases 7847
Southeast Asia, East Asia, and Oceania	Stroke 28316.5	IHD 18790.1	COPD 15783.7	Tracheal, bronchus, and lung cancer 7406.2	AD 4536.5	Stroke 19724.9	IHD 13926.1	COPD 9855.6	AD 7649.3	DM 3801.1
East Asia	Stroke 28692.8	IHD 18865.3	COPD 16612.7	Tracheal, bronchus, and lung cancer 8121.2	AD 4747	Stroke 18990	IHD 14028	COPD 10973.9	AD 8135.1	Tracheal, bronchus, and lung cancer 3456.4
Oceania	COVID-19 26899	IHD 23413.7	Stroke 23258.8	COPD 21238.2	DM 19406.8	Stroke 24894.4	IHD 19457.4	COPD 18174.1	DM 16551.3	COVID-19 13632.3
Southeast Asia	Stroke 26722.6	IHD 18409.9	COVID-19 17112.6	COPD 12091	DM 7421.9	Stroke 22554.1	IHD 13484.5	COVID-19 9153.3	DM 7838.6	AD 5774.1
Sub-Saharan Africa	COVID-19 45397.7	Stroke 20150.5	IHD 15288.6	Lower respiratory infections 10828.5	DM 9163	COVID-19 25588.9	Stroke 19900.6	IHD 12713.7	Lower respiratory infections 8420.7	DM 8090.7
Central Sub-Saharan Africa	COVID-19 45603.5	Stroke 23175	IHD 21018.1	Lower respiratory infections 16211.7	Tuberculosis 13084.5	COVID-19 24496.4	Stroke 23285.8	IHD 15796.7	HHD 13854.5	Lower respiratory infections 11269.6
Eastern Sub-Saharan Africa	COVID-19 52440.9	Stroke 20085.3	IHD 12613.7	Tuberculosis 11957.6	Lower respiratory infections 11338.1	COVID-19 28880.4	Stroke 18930.8	IHD 9343.1	HHD 9029.9	Lower respiratory infections 8886.4
Southern Sub-Saharan Africa	COVID-19 72475.7	Stroke 16029.3	IHD 13162.5	DM 12735	COPD 10027.4	COVID-19 40306.3	Stroke 18492	DM 14382.9	IHD 10451.9	HHD 7640.7
Western Sub-Saharan Africa	COVID-19 32707.2	Stroke 20681.4	IHD 16893.8	Lower respiratory infections 9989.2	DM 8146.1	Stroke 20401.2	COVID-19 17238.8	IHD 15739.8	Lower respiratory infections 8120.1	DM 7301.3

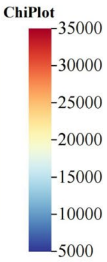

Fig. 4 Five leading causes of total DALYs rates in 2021 by region and sex for adults aged ≥ 70. *Abbreviations:* DALY = disability-adjusted life-year; AD, Alzheimer's disease and other dementias; COPD, chronic obstructive pulmonary disease; CKD, chronic kidney disease; DM, diabetes mellitus; HHD, hypertensive heart disease; IHD, ischemic heart disease

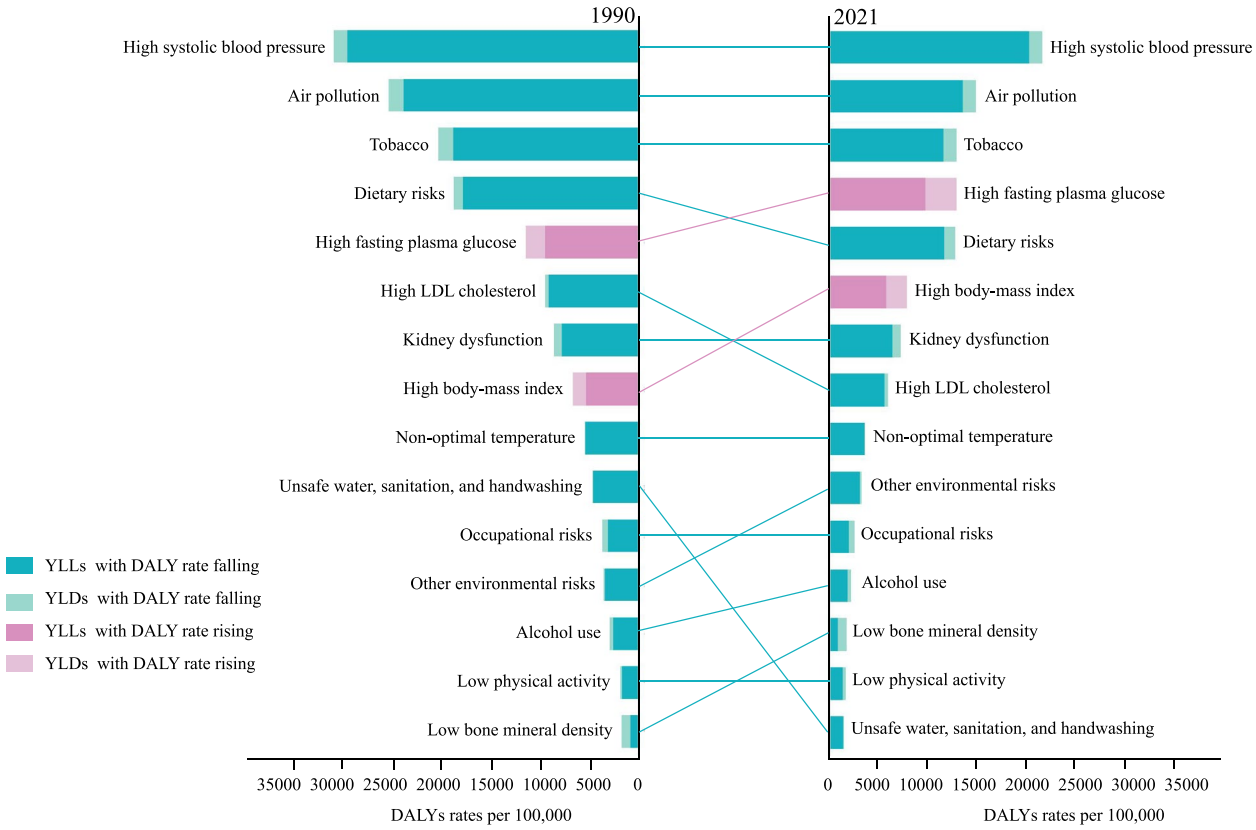

Fig. 5 Temporal changes in the top 15 risk factors of total DALYs rates among adults aged ≥ 70 between 1990 and 2021. This figure illustrates the temporal shifts in the top 15 risk factors contributing to DALY rates among adults aged 70 and older globally, comparing data from 1990 and 2021. The list of risk factors is ranked based on their DALY rates per 100,000 population in these respective years. Bars and lines are color-coded to indicate whether DALY rates were higher in 2021 (pink) or 1990 (green), determined by whether the 95% uncertainty interval of the absolute difference in DALY rates includes zero. The transparency of each bar and line reflects the composition of DALYs between Years of Life Lost (YLL) and Years Lived with Disability (YLD) for each risk factor. *Abbreviations:* DALY, disability-adjusted life-year; YLD, years lived with disability; YLL, years of life lost

DALYs rates for many risk factors have decreased from 1990 to 2021, there has been a noticeable increase in DALYs attributed to metabolic factors. This rise underscores the growing health burden posed by conditions like diabetes and obesity in aging populations diabetes and obesity in aging populations [45–47]. Policymakers can use these insights to develop interventions for better older health outcomes. Strategies include promoting healthier lifestyles through education and community programs, improving access to preventive healthcare, and addressing socio-economic disparities that worsen health inequalities among older adults worldwide. These actions aim to mitigate the impact of rising metabolic health challenges and support healthier aging across diverse socio-demographic contexts.

Strengths and limitations of the study

The current study offers comprehensive global insights into the distribution, trends, and risk factors of disease burden among adults aged 70 and over. It meticulously examines variations across sex, regional classifications, and detailed demographic categories, providing valuable evidence for understanding the evolving health challenges faced by this population. The findings have significant public health implications, as they can inform the development of targeted interventions and policies aimed at mitigating the growing burden of disease in older adults. For instance, the identification of leading risk factors such as high blood pressure, air pollution, and tobacco use can guide policymakers in prioritizing health promotion efforts and resource allocation to reduce these modified risks. Moreover, these insights can help shape healthcare systems that are better equipped to address the unique needs of older populations, such as the integration of age-specific healthcare services and the emphasis on chronic disease management.

However, it is essential to acknowledge several limitations inherent in the study. Firstly, like other research

	Male					Female				
	1	2	3	4	5	1	2	3	4	5
Global	High SBP 23307.2	Tobacco 21914.6	Air pollution 17524.8	Dietary risks 14665.8	High FPB 14029.2	High SBP 19927.1	Air pollution 12513.4	High FPB 11777.4	Dietary risks 11057.8	High BMI 8179
Central Europe, Eastern Europe, and Central Asia	High SBP 39389.7	Dietary risks 26367.7	Tobacco 22681.5	High LDL-C 15481.7	High FPB 15000.6	High SBP 34076.5	Dietary risks 19939.4	High BMI 13729.5	High FPB 13346.8	High LDL-C 13082
Central Asia	High SBP 48931.4	Dietary risks 34082.8	Tobacco 24523.5	Air pollution 22968.8	Kidney dysfunction 16168.9	High SBP 41304.3	Dietary risks 25281.2	Air pollution 17204.7	High FPB 14316.4	High BMI 14222.4
Central Europe	High SBP 32335.9	Dietary risks 23847.9	Tobacco 19232.1	High FPB 16674.5	High BMI 12927.6	High SBP 27875.5	Dietary risks 17777.8	High FPB 14273.2	High BMI 13197.6	High LDL-C 9202.3
Eastern Europe	High SBP 43577	Dietary risks 27035.4	Tobacco 25272.2	High LDL-C 18970	High FPB 13425.3	High SBP 36895.5	Dietary risks 20515.6	High LDL-C 15557.8	High BMI 13989.6	High FPB 12630.5
High-income	Tobacco 11781.5	High SBP 11437.4	High FPB 11190.3	Dietary risks 8984.1	High BMI 7609.6	High SBP 10011.6	High FPB 8796.3	High BMI 7500.6	Dietary risks 6625.8	Kidney dysfunction 4801.8
Australasia	High SBP 8757.4	High FPB 8299.1	High BMI 7223	Dietary risks 6977.9	Tobacco 5895.4	High SBP 7591.6	High BMI 7387.7	High FPB 6590	Dietary risks 5398.3	Kidney dysfunction 3827.5
High-income Asia Pacific	Tobacco 12672.9	High FPB 9238.5	High SBP 8856.9	Dietary risks 6641.3	Kidney dysfunction 4745.8	High FPB 7185.5	High SBP 6519.3	Dietary risks 4745.6	High BMI 3357	Kidney dysfunction 3312.2
High-income North America	High FPB 15819.6	High SBP 12449.6	Tobacco 12194.6	High BMI 11676.9	Dietary risks 10959	High FPB 11774.5	High SBP 11768.3	High BMI 10381.4	Dietary risks 7734.9	Tobacco 7302.8
Southern Latin America	High SBP 15461	High FPB 12378.6	Dietary risks 12129.9	High BMI 10177.5	Tobacco 9055.6	High SBP 12395.3	High BMI 10246	High FPB 9909	Dietary risks 9040.9	Kidney dysfunction 5004.4
Western Europe	High SBP 11945.8	Tobacco 11604.9	High FPB 9185.7	Dietary risks 8753.6	High BMI 7045.7	High SBP 10669.8	High FPB 7747.5	High BMI 7626.6	Dietary risks 6771.8	Kidney dysfunction 4517.4
Latin America and Caribbean	High SBP 17091.1	High FPB 16054.2	Dietary risks 11132.9	High BMI 10619.3	Tobacco 9468.4	High FPB 14669	High SBP 13821.2	High BMI 11142.1	Dietary risks 8725	Kidney dysfunction 6771.2
Andean Latin America	High FPB 13314.3	High SBP 12135.3	Dietary risks 9321.1	High BMI 8548.1	Air pollution 8407.9	High FPB 12999.2	High SBP 11563.8	High BMI 9619.6	Dietary risks 8352.4	Air pollution 7425.2
Caribbean	High SBP 21209.1	High FPB 17845.2	Dietary risks 12651.7	Tobacco 12101.5	Air pollution 10623.6	High SBP 17825.8	High FPB 17049.6	Dietary risks 10935.8	High BMI 10778.4	Air pollution 9085.3
Central Latin America	High SBP 18669.8	High FPB 18283.3	Dietary risks 12892.9	High BMI 12232.3	Kidney dysfunction 10779.6	High FPB 17215	High SBP 15509.2	High BMI 13020.9	Dietary risks 10004.1	Kidney dysfunction 8950.8
Tropical Latin America	High SBP 15771.2	High FPB 14075.3	Tobacco 11580.2	High BMI 9719.5	Dietary risks 9457.4	High FPB 12197.7	High SBP 11911.6	High BMI 9829.7	Dietary risks 7169.4	Kidney dysfunction 4809.8
North Africa and Middle East	High SBP 33691.8	High FPB 21889.3	Tobacco 20328.5	Air pollution 19383.2	Dietary risks 17823.1	High SBP 35627.9	High FPB 22636.1	High BMI 20792.5	Air pollution 17099.6	Dietary risks 16057.7
South Asia	Air pollution 34393.1	Tobacco 23514.7	High SBP 22742	High FPB 16902.6	Dietary risks 16445.3	Air pollution 27205.4	High SBP 20246.7	High FPB 13808.5	Dietary risks 13003.8	Unsafe water, sanitation, and handwashing 7132.7
Southeast Asia, East Asia, and Oceania	Tobacco 35311.3	High SBP 31166.8	Air pollution 24977	Dietary risks 17159.7	High FPB 12815.5	High SBP 23814.3	Air pollution 17590.2	Dietary risks 11279.3	High FPB 10918.1	Tobacco 7774.2
East Asia	Tobacco 37404.3	High SBP 30677.4	Air pollution 25849.2	Dietary risks 17281.5	High FPB 11547.2	High SBP 22507.8	Air pollution 17933.8	Dietary risks 11011.1	High FPB 9445.7	Tobacco 8253.3
Oceania	Air pollution 37789.6	High FPB 32934.3	High SBP 21746.2	Dietary risks 21577.7	Tobacco 20535.8	Air pollution 35096.6	High FPB 28192.8	High SBP 23792.9	Dietary risks 19003.2	High BMI 14541.5
Southeast Asia	High SBP 33414.3	Tobacco 26300.1	Air pollution 21014.7	High FPB 18153.8	Dietary risks 16577.7	High SBP 28916.5	High FPB 16532.2	Air pollution 16111.1	Dietary risks 12266	Kidney dysfunction 9628.7
Sub-Saharan Africa	High SBP 26687.3	Air pollution 26598.9	High FPB 19687.1	Dietary risks 14915.9	Kidney dysfunction 11641.5	High SBP 30189	Air pollution 22729.4	Dietary risks 16215.4	High FPB 15927	High BMI 11438.1
Central Sub-Saharan Africa	Air pollution 33709.3	High SBP 33193.4	High FPB 27726.8	Dietary risks 19710.9	Kidney dysfunction 16288.3	High SBP 39105.4	Air pollution 28939.9	Dietary risks 21304	High FPB 17690	High BMI 12734.4
Eastern Sub-Saharan Africa	Air pollution 28103.8	High SBP 22393.3	High FPB 18067	Dietary risks 15957.3	Kidney dysfunction 9930.8	High SBP 27264	Air pollution 22950	Dietary risks 17463.8	High FPB 12333.3	Kidney dysfunction 7826.5
Southern Sub-Saharan Africa	High SBP 25246.5	High FPB 22326.8	High BMI 15430.5	Dietary risks 15372.7	Air pollution 15218.3	High SBP 28862	High FPB 23031.7	High BMI 20736.3	Dietary risks 15888.1	Air pollution 13566.4
Western Sub-Saharan Africa	High SBP 29274.7	Air pollution 26942.7	High FPB 18770.9	Dietary risks 13035.4	Kidney dysfunction 12420	High SBP 30903.1	Air pollution 24457.4	High FPB 15844.3	Dietary risks 13888.7	High BMI 10739.1

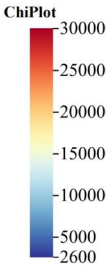

Fig. 6 Five leading risk factors of total DALYs rates in 2021 by region and sex for adults aged ≥ 70. *Abbreviations:* DALYs, deaths and disability-adjusted life years; FPG, fasting plasma glucose; LDL-C, low-density lipoprotein cholesterol; SBP: systolic blood pressure

using the GBD methodology, our study faces inherent limitations discussed extensively elsewhere [20, 47]. These include challenges in fully capturing uncertainties associated with data sources, delays in data availability, and variability in coding practices and biases. Consequently, individual risk factors may not be fully represented in population-level estimates. Second, data completeness from certain countries may be compromised due to reliance on non-representative samples. Exclusion of subnational studies focusing solely on urban areas may potentially overlook important variations by ethnicity, socioeconomic status, and within specific urban populations. This limitation underscores the need for caution when generalizing findings across diverse geographic and demographic contexts.

Conclusions

The global health burden on individuals aged 70 and older has shown a decline in both death rates and DALYs, although it remains significant, particularly due to population aging, especially in less developed regions, especially in developing regions. Effective preventive strategies are crucial for addressing key risk factors such as high systolic blood pressure, air pollution, and tobacco use. Furthermore, improving treatment options for prevalent conditions like ischemic heart disease, stroke, COVID-19, and chronic obstructive pulmonary disease is essential to effectively manage and mitigate these health challenges across populations.

Abbreviations

GBD	Global Burden of Disease
DALYs	Disability-adjusted life years
YLLs	Years of life lost
YLDs	Years lived with disability
SDI	Socio-demographic index
UIs	Uncertainty intervals
CIs	Confidence intervals
COPD	Chronic obstructive pulmonary disease

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12877-025-06095-1.

Supplementary Material 1. Fig. 1. Temporal trends of deaths and DALYs of adults aged ≥ 70 from 1990 to 2021: (A) Death numbers and rates; (B) DALYs numbers and rates. This bar and line chart depicts the annual number and rate of deaths/DALYs among adults aged 70 and older globally from 1990 to 2021. The bar chart shows the annual number of deaths/DALYs, with error bars representing the 95% uncertainty intervals. The line chart illustrates the annual death rate per 100,000 population, with shaded areas indicating the 95% uncertainty intervals for the rates. *Abbreviations:* DALYs, deaths and disability-adjusted life years. Fig. 2. Association between DALYs rates and SDI among adults aged ≥ 70 years: (A) DALY rates vs. SDI by GBD region, 1990–2021; (B) DALY rates vs. SDI across 204 countries and territories, 2021. *Abbreviations:* DALYs, deaths and disability-adjusted life years; SDI, Socio-demographic Index; GBD, Global Burden of Disease Study. Fig. 3. Relationship between SEVs and SDI for overall, and detailed risk factors by 21 GBD regions from 1990 to 2021: (A) environmental/occupational risk; (B) behavioral risk; (C) metabolic risk factors. *Abbreviations:* SDI, socio-demographic index; GBD, Global Burden of Disease Study; SEV, summary exposure value.

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Clinical trial registration

Not applicable.

Patient and Public Involvement statement:

It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

Authors' contributions

Conceptualization: X-D Zhou and W Song; Data curation and formal analysis: X-D Zhou, Q-F Chen; Writing, review, and editing: Q-F Chen, C Ni, YL Jiang, LF Chen, HT Liao, JQ Gao, SP Pan, XQ Luan, YL Wu, X-D Zhou, W Song. All authors have approved the final version of the manuscript, including the authorship list and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Data availability

All data are available in the manuscript or the supplementary materials and can also be accessed via the Global Burden of Disease (GBD) Results Tool at https://vizhub.healthdata.org/gbd-results/.

Declarations

Ethics approval and consent to participate

This study utilized publicly available data from the Global Burden of Disease (GBD) Study 2021, which is a global health resource that aggregates data from various sources under ethical standards. As the data is anonymized and does not involve direct human participation, no individual ethics approval was required. The GBD study has received ethical approval from relevant ethics committees in the countries where data collection took place, as per the guidelines of the respective institutions.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

[1]Medical Care Center, the First Affiliated Hospital of Wenzhou Medical University, Wenzhou 325035, Zhejiang, China. [2]The Center for Geriatric Medicine, Institute of Aging, Key Laboratory of Alzheimer's Disease of Zhejiang Province, the First Affiliated Hospital of Wenzhou Medical University, Wenzhou 325035, Zhejiang, China. [3]Oujiang Laboratory (Zhejiang Lab for Regenerative Medicine, Vision and Brain Health), Longwan District, 999 Jinshi Road, Yongzhong StreetZhejiang Province, Wenzhou 325035, China. [4]MAFLD Research Center, Department of Hepatology, the First Affiliated Hospital of Wenzhou Medical University, Wenzhou 325000, China.

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References

1. Hu FB. Diet strategies for promoting healthy aging and longevity: an epidemiological perspective. J Intern Med. 2024;295(4):508–31.
2. Chen QF, Chen L, Katsouras CS, et al. Global burden of heart failure and its underlying causes in 204 countries and territories, 1990–2021. Eur Heart J Qual Care Clin Outcomes; 2025. Epub ahead of print. https://doi.org/10.1093/ehjqcco/qcae110.
3. Zhang Y, Chen H, Li R, Sterling K, Song W. Amyloid beta-based therapy for Alzheimer's disease: challenges, successes and future. Signal Transduct Target Ther. 2023;8(1):248.
4. Chen QF, Shi S, Wang YF, et al. Global, regional, and national burden of valvular heart disease, 1990 to 2021. J Am Heart Assoc. 2024;13:e037991.
5. Larsen B, Sánchez-Triana E. Global health burden and cost of lead exposure in children and adults: a health impact and economic modelling analysis. Lancet Planet Health. 2023;7(10):e831–40.
6. Ijaz N, Buta B, Xue QL, et al. Interventions for frailty among older adults with cardiovascular disease: JACC state-of-the-art review. J Am Coll Cardiol. 2022;79(5):482–503.
7. Schmanske N, Ngo JM, Kalra K, Nanna MG, Damluji AA. Healthy ageing in older adults with cardiovascular disease. Eur Heart J. 2025.
8. The Lancet Healthy L. The decade of healthy ageing: progress and challenges ahead. Lancet Healthy Longev. 2024;5(1):e1.
9. de Azeredo Passos VM, Champs APS, Teixeira R, et al. The burden of disease among Brazilian older adults and the challenge for health policies: results of the global burden of disease study 2017. Popul Health Metr. 2020;18(Suppl 1):14.
10. Al-Hajj S, Farran S, Sibai AM, et al. Injury burden in individuals aged 50 years or older in the Eastern Mediterranean region, 1990–2019: a systematic analysis from the global burden of disease study 2019. Lancet Healthy Longev. 2022;3(4):e253–62.
11. Molassiotis A, Kwok SWH, Leung AYM, Tyrovolas S. Associations between sociodemographic factors, health spending, disease burden, and life expectancy of older adults (70 + years old) in 22 countries in the Western Pacific Region, 1995–2019: estimates from the Global Burden of Disease (GBD) study 2019. Geroscience. 2022;44(2):925–51.
12. Keating N. A research framework for the United Nations decade of healthy ageing (2021–2030). Eur J Ageing. 2022;19(3):775–87.
13. GBD 2019 Ageing Collaborators. Global, regional, and national burden of diseases and injuries for adults 70 years and older: systematic analysis for the global burden of disease 2019 study. BMJ. 2022;376:e068208.
14. Chen X, Giles J, Yao Y, et al. The path to healthy ageing in China: a Peking University-Lancet Commission. Lancet. 2022;400(10367):1967–2006.
15. Rudnicka E, Napierała P, Podfigurna A, Męczekalski B, Smolarczyk R, Grymowicz M. The World Health Organization (WHO) approach to healthy ageing. Maturitas. 2020;139:6–11.
16. Pit S, Fisk M, Freihaut W, et al. COVID-19 and the ageing workforce: global perspectives on needs and solutions across 15 countries. Int J Equity Health. 2021;20(1):221.
17. Zou Z, Cini K, Dong B, et al. Time trends in cardiovascular disease mortality across the BRICS: an age-period-cohort analysis of key nations with emerging economies using the global burden of disease study 2017. Circulation. 2020;141(10):790–9.
18. Murray CJL. Findings from the global burden of disease study 2021. Lancet. 2024;403(10440):2259–62.
19. Global incidence, prevalence, years lived with disability (YLDs), disability-adjusted life-years (DALYS), and healthy life expectancy (hale) for 371 diseases and injuries in 204 countries and territories and 811 subnational locations, 1990-2021: a systematic analysis for the global burden of disease study 2021. Lancet 2024;403(10440):2133–61.
20. Global burden and strength of evidence for 88 risk factors in 204 countries and 811 subnational locations, 1990–2021: a systematic analysis for the global burden of disease study 2021. Lancet. 2024;403(10440):2162–203.
21. Burden of disease scenarios for 204 countries and territories, 2022–2050: a forecasting analysis for the global burden of disease study 2021. Lancet. 2024;403(10440):2204–56.
22. Hu J, Zhou R, Ding R, Ye DW, Su Y. Effect of PM(2.5) air pollution on the global burden of lower respiratory infections, 1990–2019: a systematic analysis from the global burden of disease study 2019. J Hazard Mater. 2023;459: 132215.
23. Diabetes mortality and trends before 25 years of age: an analysis of the global burden of disease study 2019. Lancet Diabetes Endocrinol. 2022;10(3):177–92.
24. Tong M, Lu H, Xu H, et al. Reduced human fecundity attributable to ambient fine particles in low- and middle-income countries. Environ Int. 2024;189: 108784.
25. Pu L, Wang L, Zhang R, Zhao T, Jiang Y, Han L. Projected global trends in ischemic stroke incidence, deaths and disability-adjusted life years from 2020 to 2030. Stroke. 2023;54(5):1330–9.
26. Zhou XD, Targher G, Byrne CD, et al. An international multidisciplinary consensus statement on MAFLD and the risk of CVD. Hepatol Int. 2023;17(4):773–91.
27. Roth GA, Mensah GA, Johnson CO, et al. Global burden of cardiovascular diseases and risk factors, 1990–2019: update from the GBD 2019 study. J Am Coll Cardiol. 2020;76(25):2982–3021.
28. Zhang S, Zhao J, Zhang Y, et al. Upregulation of MIF as a defense mechanism and a biomarker of Alzheimer's disease. Alzheimers Res Ther. 2019;11(1):54.
29. Schiffrin EL, Fisher NDL. Diagnosis and management of resistant hypertension. BMJ. 2024;385: e079108.
30. Sicard P, Agathokleous E, Anenberg SC, De Marco A, Paoletti E, Calatayud V. Trends in urban air pollution over the last two decades: a global perspective. Sci Total Environ. 2023;858(Pt 2):160064.
31. Spatial, temporal, and demographic patterns in prevalence of smoking tobacco use and attributable disease burden in 204 countries and territories, 1990–2019: a systematic analysis from the global burden of disease study 2019. Lancet. 2021;397(10292):2337–60.
32. Wu X, Zhu B, Xu S, Bi Y, Liu Y, Shi J. A cross country comparison for the burden of cardiovascular disease attributable to tobacco exposure in China, Japan, USA and world. BMC Public Health. 2020;20(1):888.
33. Tamil Selvan S, Yeo XX, van der Eijk Y. Which countries are ready for a tobacco endgame? A scoping review and cluster analysis. Lancet Glob Health. 2024;12(6):e1049–58.
34. Streck JM, Rigotti NA, Livingstone-Banks J, et al. Interventions for smoking cessation in hospitalised patients. Cochrane Database Syst Rev. 2024;5(5):Cd001837.
35. Zhang H, Zhou XD, Shapiro MD, et al. Global burden of metabolic diseases, 1990–2021. Metabolism. 2024;160:155999. Epub 2024 Aug 14. https://doi.org/10.1016/j.metabol.2024.155999.
36. Liu J, Mozaffarian D. Trends in diet quality among U.S. adults from 1999 to 2020 by race, ethnicity, and socioeconomic disadvantage. Ann Intern Med. 2024;177(7):841–50.
37. Smith HJ, Sharma A, Mair WB. Metabolic communication and healthy aging: where should we focus our energy? Dev Cell. 2020;54(2):196–211.
38. Li Y, Wang K, Jigeer G, et al. Healthy lifestyle and the likelihood of becoming a centenarian. JAMA Netw Open. 2024;7(6): e2417931.
39. Zhang YB, Chen C, Pan XF, et al. Associations of healthy lifestyle and socioeconomic status with mortality and incident cardiovascular disease: two prospective cohort studies. BMJ. 2021;373: n604.
40. Wang J, Chen C, Zhou J, et al. Healthy lifestyle in late-life, longevity genes, and life expectancy among older adults: a 20-year, population-based, prospective cohort study. Lancet Healthy Longev. 2023;4(10):e535–43.
41. Miller MR, Landrigan PJ, Arora M, Newby DE, Münzel T, Kovacic JC. Environmentally not so friendly: global warming, air pollution, and wildfires: JACC focus seminar, part 1. J Am Coll Cardiol. 2024;83(23):2291–307.
42. Sultan Y, Salman Z, Alzaatreh M, et al. Smoking-related disease impact in the Eastern Mediterranean region: a comprehensive assessment using global burden of disease data. Asian Pac J Cancer Prev. 2024;25(2):495–505.

43. Alhuneafat L, Ta'ani OA, Jabri A, et al. Cardiovascular disease burden in the Middle East and North Africa region. Curr Probl Cardiol. 2024;49(3):102341.
44. Zhou XD, Chen QF, Targher G, et al. Global burden of disease attributable to metabolic risk factors in adolescents and young adults aged 15–39, 1990–2021. Clin Nutr. 2024;43(12):391–404.
45. Dong XJ, Zhang XQ, Wang BB, Hou FF, Jiao Y, Wang JG. The burden of cardiovascular disease attributable to high body mass index-an observational study. Eur Heart J Qual Care Clin Outcomes. 2024;10(2):154–67.
46. Zhou XD, Chen QF, Yang W, et al. Burden of disease attributable to high body mass index: an analysis of data from the global burden of disease study 2021. eClinicalMedicine. 2024;76:102848.
47. Dai H, Alsalhe TA, Chalghaf N, Riccò M, Bragazzi NL, Wu J. The global burden of disease attributable to high body mass index in 195 countries and territories, 1990–2017: an analysis of the global burden of disease study. PLoS Med. 2020;17(7): e1003198.

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